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BECE

BASIC DESIGN AND TECHNOLOGY

(Pre-Technical Skills)

Past Questions & Answers

2010 - 2017

Junior High School
Years 1, 2 & 3

Compiled by:

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- **Teachers, parents and educational institutions** – for your continuous patronage and tireless efforts at ensuring quality education for the 21st century child
- Junior High School **students** – you are the ultimate reason for this work.

DEDICATION

To the Lord **Jesus Christ**, our Saviour and soon-coming King

LIST OF PAPERS (Hold down **Ctrl & Click** on a link to get there)

2017	**PRE-TECH**	**PAPER 1**
2017	**PRE-TECH**	**PAPER 2**
2016	**PRE-TECH**	**PAPER 1**
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2015	**PRE-TECH**	**PAPER 1**
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2010	**PRE-TECH**	**PAPER 1**
2010	**PRE-TECH**	**PAPER 2**

June 2017

BASIC DESIGN AND TECHNOLOGY 1
(Pre-Technical Skills 1)

Objective Test
40 minutes

1. A **common** method of cooking ripe plantain for sale is
A. boiling.
B. roasting.
C. grilling.
D. baking.

2. To shorten the cooking time for meat,
A. slice along the grain.
B. cut into small pieces.
C. slice into thin pieces.
D. cut across the grain.

3. In taking body measurement, a firm tape measure is needed to
A. save time.
B. avoid wasting fabric.
C. obtain accuracy.
D. make sewing easy.

4. Which of the following principles of design is similar to proportion?
A. Balance
B. Perspective
C. Rhythm
D. Scale

5. A preliminary design is meant to
A. improve creativity.
B. increase production.
C. make sketches.
D. produce ideas.

Use the sketch in Figure 1 to answer Questions 6 and 7

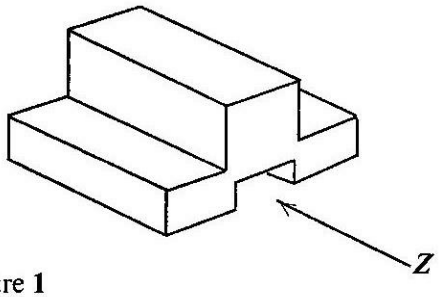
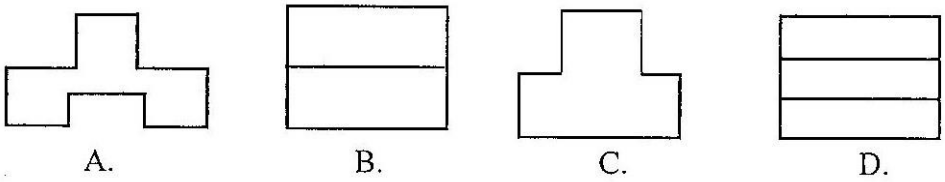
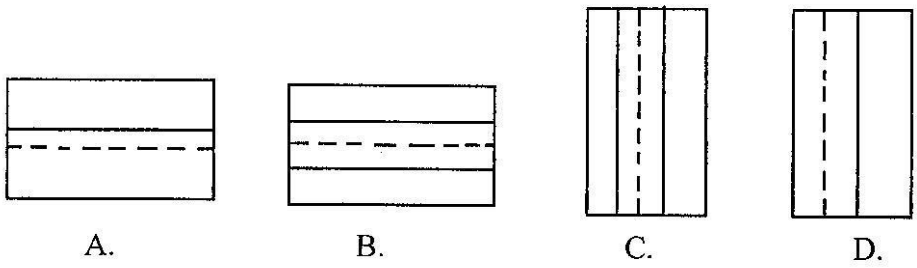


Figure 1

6. Which of the following represents the front view in the direction of arrow Z?



7. Which of the following represents the left end elevation?



8. A single unit of a design is known as

A. motif.
B. pattern.
C. plan.
D. print.

9. Angles are measured by using a

A. compass.
B. protractor.
C. rule.
D. set square.

10. In designing, the analysis question, what is the purpose of the unit? falls under

A. construction.
B. ergonomics.
C. function.
D. material.

11. To repair an electrical fault, a safety precaution to observe is to

A. disconnect live wires.
B. remove fuse.
C. remove cover of socket.
D. switch off main supply.

12. Which of the following is **not** a cause of weakness in furniture?
A. Loose joint
B. Insufficient polish
C. Overloading
D. Removal of nail

13. The tee-square is used for drawing
A. arcs.
B. circles.
C. horizontal lines.
D. vertical lines.

14. Identify the type of lettering used for emphasis.
A. Freestyle
B. Gothic
C. Italics
D. Romans

15. Strips cut on the true cross of fabric are usually used for curved edges because they
A. stretch easily.
B. do not fray.
C. are easily prepared.
D. are colourful.

16. The body of a box-iron is made from
A. aluminium.
B. cast iron.
C. copper.
D. wrought iron.

17. Polyvinyl acetate is a type of
A. abrasive.
B. adhesive.
C. aggregate.
D. finish.

18. The type of aggregate that passes through a 5 mm wire mesh is
A. clay.
B. laterite.
C. sand.
D. stone.

19. Which of the following tools is suitable for pairing on wood?

A. Cross-cut saw
B. Firmer chisel
C. Mortise chisel
D. Rip saw

Use the sketch in Figure **2** *to answer questions* **20** *and* **21**.

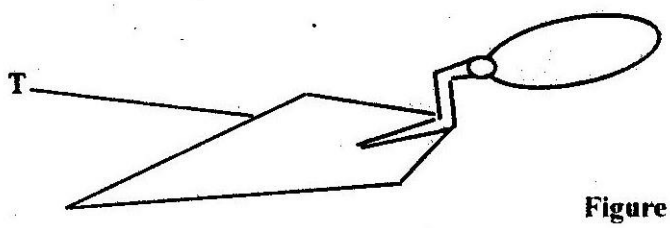


Figure 2

20. The tool is used for
A. cutting bricks.
B. mixing mortar.
C. picking and spreading mortar.
D. setting out a wall.

21. The part labelled T is
A. blade.
B. ferrule.
C. handle.
D. shank.

22. Personal safety is observed on site in order to
A. reduce the risk of damaging tools
B. prevent damage to workpiece.
C. care and maintain tools.
D. prevent accidents to workers.

Use the sketch in Figure **3** *to answer questions* **23** *and* **24**

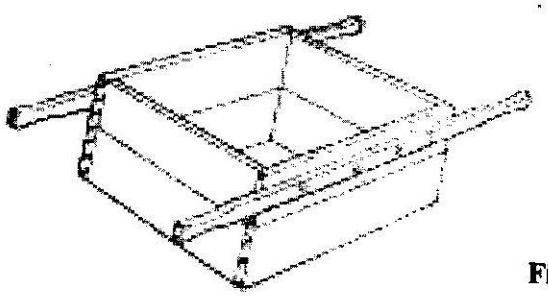


Figure 3

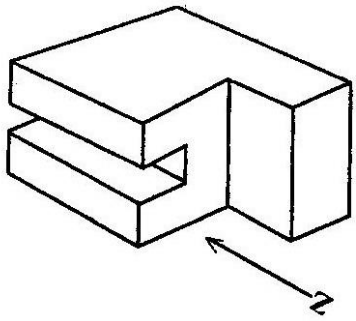
23. The sketch shown is
A. gauge box.
B. head pan.
C. mould box.
D. pallet.

24. The tool is used for

A. holding blocks.
B. moulding blocks.
C. fetching water.
D. batching materials.

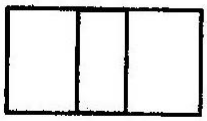
Use the sketch in Figure **4** *to answer questions* **25** *and* **26**

Figure 4

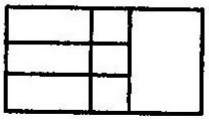


25. Which of the following represents the front view in the direction of arrow Z?

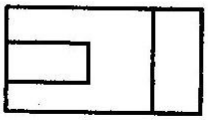
A.



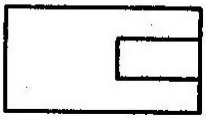
B.



C.



D.



26. Which of the following represents the left end elevation?

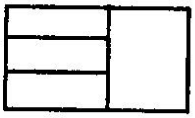
A.



B.



C.



D.



Use the sketch in Figure **5** *to answer questions* **27** *and* **28**

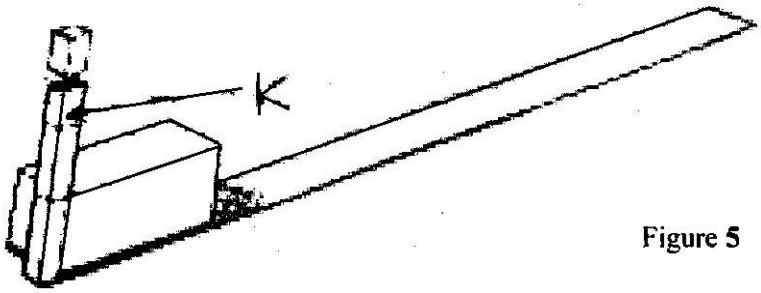


Figure 5

27. The sketch shows the process of

A. plumbing a wall.
B. levelling a wall.
C. spreading mortar for a wall.

D. checking height of a wall.

28. The tool labelled K is the

A. builder's square.
B. gauge rod.
C. spirit level.
D. straight edge.

29. In an electrical circuit, the fuse is connected to the

A. earth wire.
B. live wire.
C. negative wire.
D. neutral wire.

30. Which of the following components helps to open and close a circuit?

A. Cell
B. Fuse
C. Plug
D. Switch.

June 2017

BASIC DESIGN AND TECHNOLOGY 1
(Pre-Technical Skills 1)
Objective Test
ANSWERS

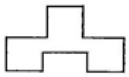
1. C. grilling

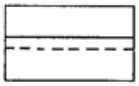
2. D. cut across the grain

3. C. obtain accuracy

4. D. Scale

5. D. produce ideas

6. A. 

7. A 

8. A. motif

9. B. protractor

10. C. function

11. D. switch off main supply

12. B. Insufficient polish

13. C. horizontal lines

14. C. Italics

15. A. stretch easily

16. B. cast iron

17. B. adhesive

18. C. sand

19. C. Mortise chisel

20. C. picking and spreading mortar

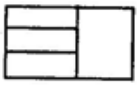
21. A. blade

22. D. prevent accidents to workers

23. A. gauge box

24. D. batching materials

25. C. 

26. C. 

27. A. plumbing a wall

28. C. spirit level

29. B. live wire

30. D. Switch

June 2017

BASIC DESIGN AND TECHNOLOGY 2
(Pre-Technical Skills 2)

Essay
1 hour 15 minutes

SECTION A
(CORE SKILLS)
[20 marks]
Answer question 1 [COMPULSORY]

1. (a) What are *temporary stitches*? [2 marks]

(b) State **four** reasons for setting a table for family meals. [4 marks]

(c) A unit for hanging school uniforms is to be designed and made in the school workshop.

(i) State **two** places where the unit would be placed in the workshop. [3 marks]

(ii) State **two** specifications of the unit based on construction; [2 marks]

(iii) Name **one** specific material to be used for making the unit. [2 marks]

(d) (i) List **five** important pieces of information printed onto a package. [5 marks]

(ii) Name **two** other methods for making a print. [2 marks]

SECTION B
[50 marks]

Answer **two** *questions* **only** *from this section*
All questions carry equal marks

2. (a) (i) Make a freehand pictorial sketch of the hand file with a handle used for filing metals.

(ii) Label any **two** parts of the hand file sketched in question 2 (a)(i).

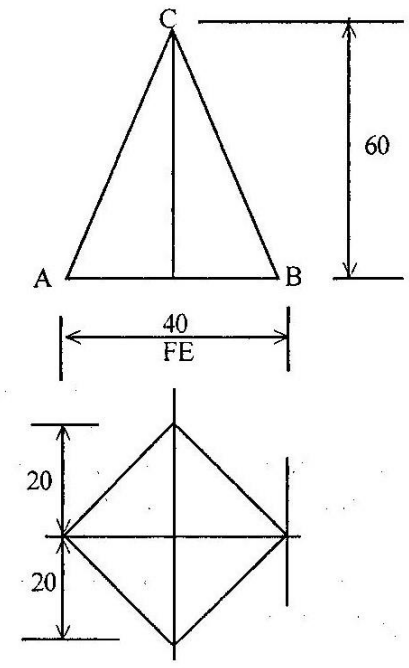
(iii) State **one** use of the hand file.

(b) Copy and complete the table below:

OPERATION	TWO TOOLS REQUIRED FOR EACH OPERATION
(i) Moulding of bricks	
(ii) Marking out on metal plate	
(iii) Soldering sheet metals	
(iv) Cutting a piece of wood into two	

Figure 1 shows the front view and plan of a square pyramid.

Figure 1



PLAN

(c) Draw full size the following views:
(i) front view;
(ii) plan.

[25 marks]

3. (a) (i) Make a freehand pictorial sketch of the claw hammer.

(ii) Label any **two** parts of the claw hammer sketched in question 3(a)(i)

(iii) State **one** use of the claw hammer.

(b) Copy and complete the table below:

PRODUCT	**ONE** SPECIFIC MATERIAL FOR MAKING THE PRODUCT
(i) Body of sewing machine	
(ii) Water hose	
(iii) Gauge box	
(iv) Traditional oven	
(v) Drawing board	

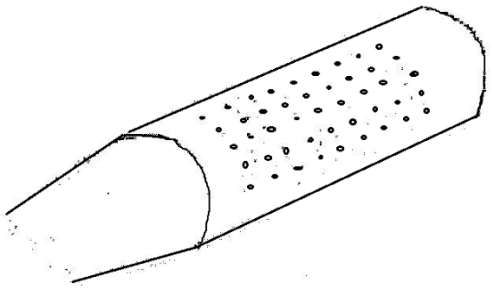


Figure 2

(c) Figure **2** shows the pictorial sketch of a cassava grater made from sheet metal.
Re-arrange the following procedure for making the grater in the correct order:

(i) cut out the shape with snips;
(ii) draw patterns on paper;
(iii) punch holes on the sheet metal;
(iv) solder closed edges;
(v) fold workpiece.

(d) (i) Explain the term batching in blockwork.

(ii) Name one tool used for mixing mortar.

[25 marks]

4. (a) (i) Make a freehand pictorial sketch of a ball-pein hammer.

(ii) Label any **two** parts of the ball-pein hammer sketched in question 4(a)(i).

(iii) State **one** use of the ball-pein hammer.

(b) Copy and complete the table below

MATERIAL	ONE USE	ONE PROPERTY
(i) Mahogany		
(ii) Nylon		
(iii) Clay		

(c) (i) List **two** tools used for reviting metal pieces.

(ii) State **two** safety precautions to be observed when building a wall.

(iii) List **two** types of chisels used in wood work.

[25 marks]

END OF ESSAY TEST

June 2017

BASIC DESIGN AND TECHNOLOGY 2
(Pre-Technical Skills 2)
Objective Test
ANSWERS

1. (a) What are *temporary stitches*? [2 marks]

These are stitches used to hold parts of articles together and removed when permanent stitches are made.

(b) State four reasons for setting a table for family meals. [4 marks]

(State any **four** of the following)

- To stimulate / whet appetite
- To have all items needed within reach
- To make eating more pleasant
- To create a welcoming environment
- It brings out the skills of the caterer
- Young ones are taught how to set table

(c) A unit for hanging school uniforms is to be designed and made in the school workshop. [7 marks]

(i) State two places where the unit would be placed in the workshop.

(State any **two** of the following)

- In the wardrobe
- On the wall
- Partition bar
- Behind the door
- On the window frame

(ii) State two specifications of the unit based on construction;

(State any **two** of the following)

- Must be strong
- Should take more uniforms at a time
- Corrosive resistant
- Use all types of joints

- Screwed
- Must be portable

(iii) Name one specific material to be used for making the unit.

(Mention any **one** of the following)

- Odum
- Sapele
- Wawa
- Raffia palm
- Cane
- PVC (acrylic)
- mild steel
- Aluminium

(d) (i) List five important pieces of information printed onto a package. [5 marks]

(Mention any **five** of the following)

- The name of the product
- Manufacturing date
- Expiry date
- Bar code
- The ingredients used
- Picture of content
- Country of origin
- Usefulness of product
- Net weight / volume of the product

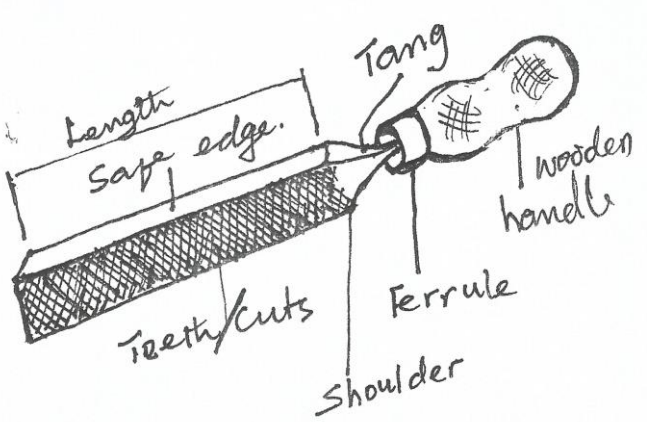
(ii) Name two other methods for making a print. [2 marks]

- Direct printing from existing surfaces, e.g. leaf, tree bark, coin
- Indirect printing, e.g., block, screen, stencil, roller, relief and frontage

SECTION B
[50 marks]

Answer **two** *questions* **only** *from this section*
All questions carry equal marks

2. (a) (i) Make a freehand pictorial sketch of the hand file with a handle used for filing metals.



(ii) Label any two parts of the hand file sketched in question 2 (a)(i).

(State any **two** of the following)
- Tang
- Blade
- Body
- Edge
- Teeth / cuts
- Handle
- Ferrule

(iii) State one use of the hand file.

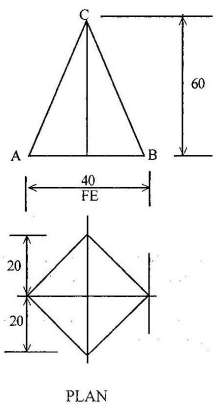
(State any **one** of the following)
- Used to file shoulders of metal
- Used to file corners of metal

(b) Copy and complete the table below:

OPERATION	TWO TOOLS REQUIRED FOR EACH OPERATION
(i) Moulding of bricks	Mould box Tamping Rod Shovel, Spade Wheel barrow / Head pan Gauge box
(ii) Marking out on metal plate	Scriber Steel Rule / Rule / Ruler Try square Dot Punch Chalk Oddleg Calipers A pair of compasses / wing compasses Pair of Dividers Tape measure Pencil
(iii) Soldering sheet metals	Soldering Bit / Stove / Soldering iron/ Soldering hammer Brush, File Tin snips / snips / shears
(iv) Cutting a piece of wood into two	Cross cut saw Rip saw Tenon saw Dovetail saw Hand saw (all specific woodwork saws)

Figure 1 shows the front view and plan of a square pyramid.

Figure 1



PLAN

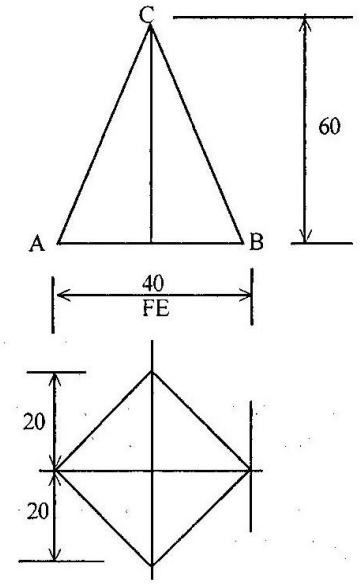
(c) Draw full size the following views:

(i) front view;

(ii) plan.

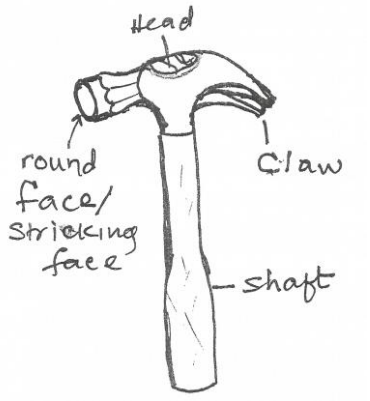
[25 marks]

Figure 1



PLAN

3. (a) (i) Make a freehand pictorial sketch of the claw hammer.



(ii) Label any **two** parts of the claw hammer sketched in question 3(a)(i)

(State any **one** of the following)

- Claw
- Head
- Shaft
- Striking face

(iii) State one use of the claw hammer. [2 marks]

(State any **one** of the following)

- For removing nails
- For driving in nails

(b) Copy and complete the table below: **[5 marks]**

PRODUCT		**ONE** SPECIFIC MATERIAL FOR MAKING THE PRODUCT
(i)	Body of sewing machine	Cast Iron
(ii)	Water hose	PVC rubber
(iii)	Gauge box	Wawa (or any specific wood/metal)
(iv)	Traditional oven	Clay laterite
(v)	Drawing board	Acrylic wawa, plywood sapele (or any specific wood mentioned)

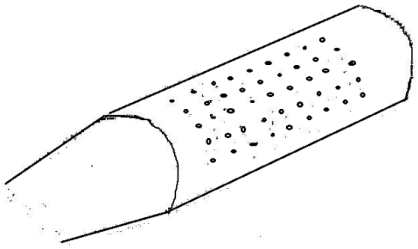


Figure 2

(c) Figure 2 shows the pictorial sketch of a cassava grater made from sheet metal. Re-arrange the following procedure for making the grater in the correct order: [2 marks]

1. Draw patterns on paper;
2. Cut out the shape with snips;
3. Punch holes on the sheet metal;
4. Fold workpiece into shape.
5. Solder closed edges;

Note: Steps 2 and 3 are interchangeable

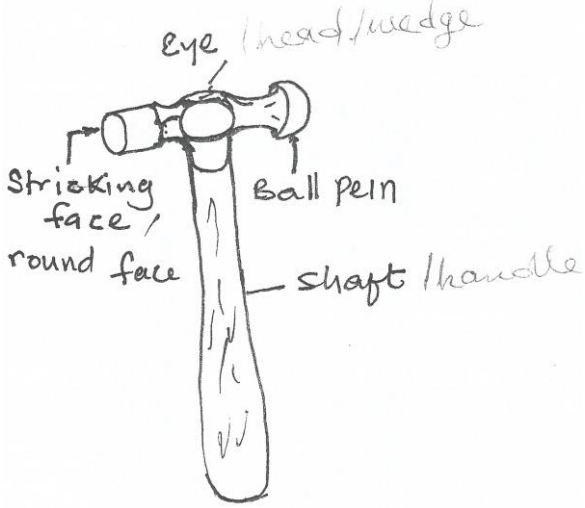
(d) (i) Explain the term batching in blockwork. **[3 marks]**

Measuring of aggregates

(ii) Name one tool used for mixing mortar. **[1 mark]**

Spade / shovel

4. (a) (i) Make a freehand pictorial sketch of a ball-pein hammer. **[6 marks]**



(ii) Label any two parts of the ball-pein hammer sketched in question 4(a)(i). **[2 marks]**

(State any **two** of the following)

- handle / shaft
- striking face
- eye / wedge / head
- ball pein
- head

(iii) State one use of the ball-pein hammer. **[2 marks]**

(State any **one** of the following)

- Used for riveting / shaping
- Used for driving nails / pins

(b) Copy and complete the table below **[7 marks]**

MATERIAL	**ONE USE** (State any **one** of the following)	**ONE PROPERTY** (State any **one** of the following)
(i) Mahogany	Panelling, Plywood, furniture	Hard, durable, mixed grain / Interlock grain, brownish / brownish red / red in colour
(ii) Nylon	Clothing, carpets, comb, brushes, gears, plug	Hard, resistance to wear
(iii) Clay	Ceramic wares, bricks, tiles, oven, water cooler, pots, mugs, bowls, earthen ware	Sticky when wet, hard when dry, dense, can hold water, prone to cracks

(c) (i) List two tools used for riveting metal pieces. **[2 marks]**

(State any **two** of the following)

- ball-pein hammer
- vice
- dolly
- rivet set

(ii) State two safety precautions to be observed when building a wall. [4 marks]

(State any **two** of the following)

- Wearing of helmet
- Wearing safety boot
- Wearing of overall / overcoat

(iii) List two types of chisels used in wood work. [2 marks]

(State any **two** of the following)

- Firmer chisel
- Mortise chisel
- Bevel edge chisel

June 2016

BASIC DESIGN AND TECHNOLOGY 1
(Pre-Technical Skills 1)

Objective Test
40 minutes

1. The basic stitch which starts **all** crocheted articles is
A. chain
B. cross
C. loop
D. stem

2. A glass is **best** used to serve
A. cream
B. drinks
C. ice-cream
D. yoghurt

3. Which of the following is inexpensive to use in the rural area?
A. Coal pot
B. Electric cooker
C. Gas cooker
D. Kerosene stove

4. The visitors book signed at exhibitions help the exhibitor to know
A. the number of people who came
B. how many works were sold
C. the number of people who showed interest
D. what people observed about the work

5. In drawing, objects nearer the viewer are in the
A. background
B. foreground
C. horizon
D. middle ground

Use the sketch in Figure **1** *to answer* Questions **6** and **7**

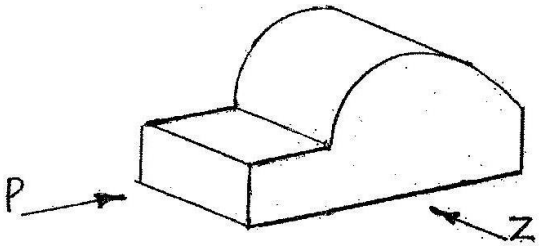
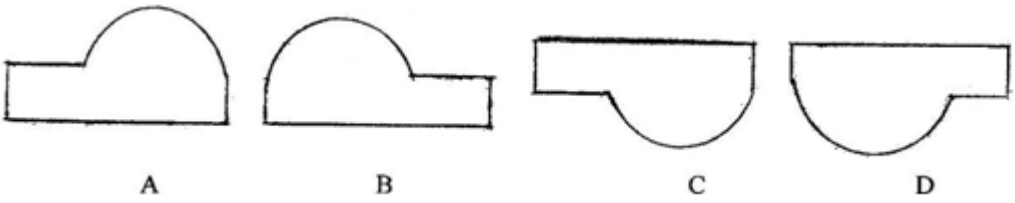
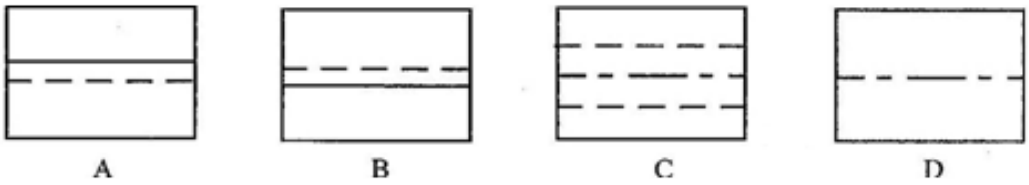


Figure 1

6. Which of the following represents the front elevation in the direction of arrow **Z**?



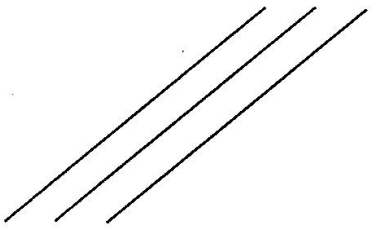
7. Which of the following represents the left end elevation in the direction of P?



8. Rhythm, balance and emphasis are

A. design processes
B. elements of art
C. idea developments
D. principles of design

9.



The lines drawn above are

A. diagonal
B. parallel
C. vertical
D. zigzag

10. The final stage of the design process is the

A. evaluation
B. possible solutions
C. specification
D. working drawings

11. Which of the following is **not** a method of recording a data for solving a design problem?

A. Observation
B. Photographs
C. Sketches
D. Write-up

12. Which of the following joints will be suitable for a chair to be dismantled and reassembled?
A. Nail lap joint
B. Glued butt joint
C. Glued screw joint
D. Bolt and nut

13.

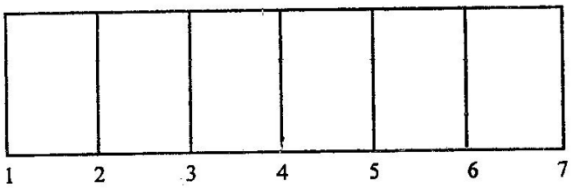


Figure 2

Figure **2** shows the surface development of a
A. cylinder
B. hexagonal prism
C. hexagonal pyramid
D. square prism

14. Select a shading style from the following options
A. Glazing
B. Hatching
C. Sgraffito
D. Sketching

15. A formal outlet for raising money to start a sewing business is
A. a family member
B. a rural bank
C. money lenders
D. the susuman

16. The tool for compacting mortar in a mould box is a
A. gauge rod
B. mallet
C. rammer
D. tamping rod

17. Which of the following tools is suitable for *pairing* in wood?
A. Cross-cut chisel
B. Firmer chisel
C. Flat chisel
D. Mortise chisel

18. The sketch in Figure **3** shows a rivet with

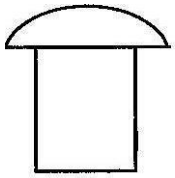


Figure 3

A. countersunk head
B. flat head
C. pan head
D. snap head

19. Which of the following plastics is **most** suitable for making insulators?
A. Acrylic
B. Polyester resin
C. Polyvinyl chloride
D. Nylon

20. The **main** difference between bricks and blocks is that
A. bricks are smaller than blocks
B. blocks are smaller than bricks
C. bricks are made of clay
D. blocks are made of laterite

21. Which of the following protective clothing is used to protect the eyes when grinding a cold chisel?
A. Gloves
B. Goggles
C. Helmet
D. Mask

Use the sketch in Figure **4** *to answer* questions **22** and **23**

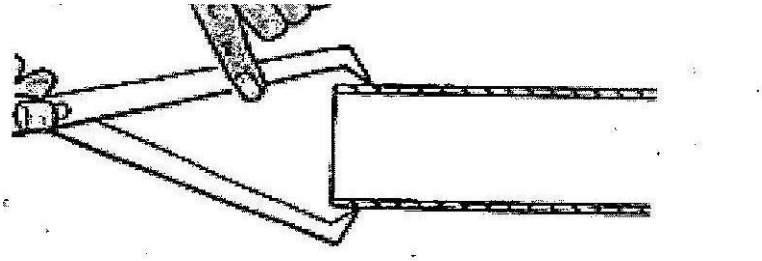


Figure 4

22. The figure shows the process of
A. checking internal diameter of a workpiece
B. checking external diameter of a workpiece.
C. drawing a circle on a workpiece
D. measuring the length of a workpiece.

23. The tool being used for the operation is
A. inside calliper
B. odd leg caliper
C. outside caliper
D. veneer caliper

Use the sketch in figure **5** *to answer* Questions **24** and **25**

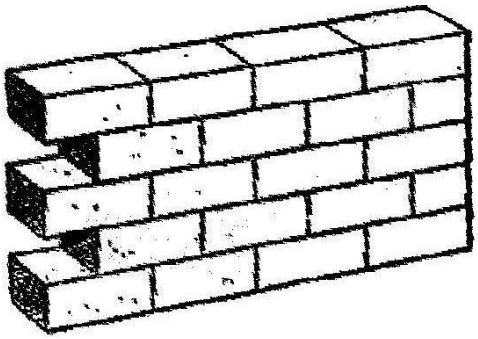


Figure 5

24. The sketch shows a

A. 4 course wall
B. 5 course wall
C. 6 course wall
D. 7 course wall

25. The method for future continuation of the wall shown is

A. block indent
B. bonding
C. racking back
D. toothing

26. Electrical parts are covered with plastic materials for

A. insulating against shock
B. outside decoration of parts
C. protection against rust
D. improvement in conductivity

27. Turpentine is a type of

A. abrasive
B. adhesive
C. aggregate
D. finish

28. The methods of drawing objects in pictorial form are

I. Isometric
II. Oblique
III. Orthographic

A. I and II only
B. I, II and III
C. I and III only
D. II and III only

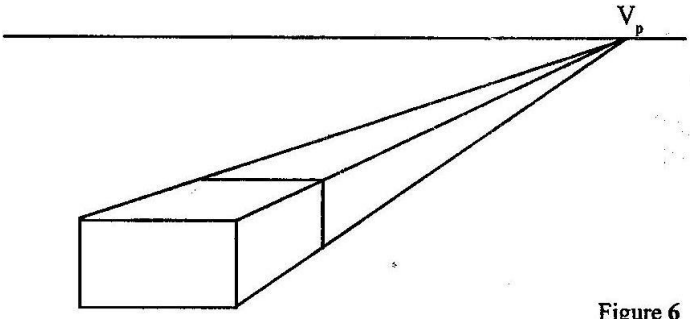


Figure 6

29. The drawing shown in Figure 6 is a/an

A. one-point perspective
B. two-point perspective
C. orthographic projection
D. isometric projection

30. The soft solder is an alloy of

A. lead and zinc
B. lead and tin
C. tin and zinc
D. zinc and copper

June 2016

BASIC DESIGN AND TECHNOLOGY 1
(Pre-Technical Skills 1)
Objective Test
ANSWERS

1.	A.	chain
2.	B.	drinks
3.	A.	Coal pot
4.	D.	what people observed about the work
5.	B.	foreground
6.	A	
7.	B	
8.	D.	principles of design
9.	B.	parallel
10.	A.	evaluation
11.	A.	Observation
12.	D.	Bolt and nut
13.	B.	hexagonal prism
14.	B.	Hatching
15.	B.	a rural bank
16.	D.	tamping rod
17.	B.	Firmer chisel
18.	D.	snap head
19.	C.	Polyvinyl chloride
20.	A.	bricks are smaller than blocks
21.	B.	Goggles
22.	B.	checking external diameter of a workpiece
23.	C.	outside caliper
24.	B.	5 course wall
25.	D.	toothing
26.	A.	insulating against shock
27.	D.	finish
28.	A.	I and II only
29.	A.	one-point perspective
30.	B.	lead and tin

June 2016

BASIC DESIGN AND TECHNOLOGY 2
(Pre-Technical Skills 2)

Essay
1 hour 15 minutes

SECTION A
(CORE SKILLS)
[20 marks]
Answer question 1 [COMPULSORY]

1. (a) List **four** stitches that can be used for neatening as well as embroidery work.
[4 marks]

(b) List **four** dishes which are easily contaminated [2 marks]

(c) (i) List **three** sources where information can be obtained to solve design problems
[3 marks]

(ii) List **two** methods of drawing objects in pictorial views. [2 marks]

(iii) List **two** causes of weaknesses in wooden chairs. [2 marks]

(d) (i) State **one** use of **each** of the elements listed below in a design

1. Shape:
2. Colour
3. Space:
4. Texture:

[4 marks]

(ii) Explain what a sketch is.
[2 marks]

(iii) State **one** reason why it is necessary to use a sketch when producing an object
[1 marks]

SECTION B
[50 marks]

Answer **two** *questions* **only** *from this section*

All questions carry equal marks

2. Figure **1** shows the front elevation and plan of an object made from a mild steel sheet.

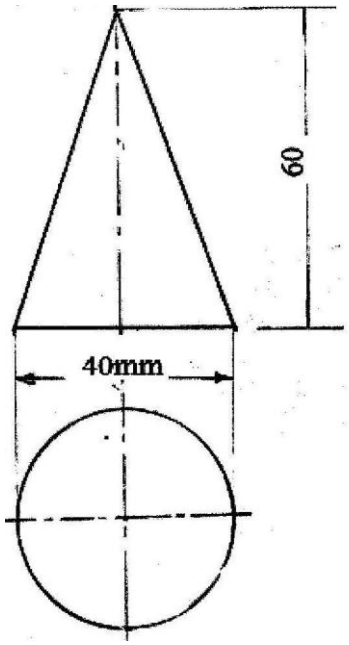


Figure 1

(a) Draw full size the following:
- (i) front elevation
- (ii) plan
- (iii) surface development

(b) (i) State the name of the object shown in Figure **1**
- (ii) List **two** tools used for cutting the mild steel sheet.

(c) (i) Make a freehand pictorial sketch of a try square.
- (ii) Indicate and label any **two** parts of the try square sketched in question **2**(c)(i)

[25 marks]

3. (a) List **two** tools **each** used in carrying out the following operations in blockwork:
- (i) laying
- (ii) setting out.

(b) (i) Make a freehand pictorial sketch of a spade
- (ii) Indicate and label any two parts of the spade sketched in question 3b(i)

(c) (i) State **one** use of the tenon saw.
- (ii) State **one** way of maintaining the tenon saw.

(d) Sketch the representative symbol for **each** of the following electrical components:
(i) switch
(ii) battery.

(e) Copy and complete the table below:

Material	*One Property*	*One use*
(i) Clay		
(ii) Wawa		
(iii) Aluminium		

[25 marks]

4. (a) Copy and complete the table below by supplying **one** tool **each** for **each** of the following operations

Operation	*Blockwork*	*Woodwork*	*Metalwork*
(i) Cutting			
(ii) Measuring			
(iii) Striking			

(b) Figure **2** shows the sketch of a handtool. *Use it to answer the following questions*:

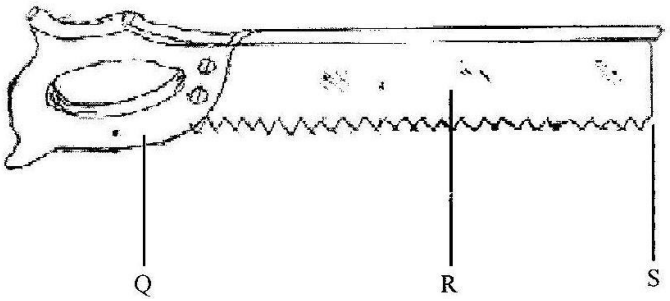


Figure **2**

(i) Write the name of the tool.
(ii) State **one** use of the tool
(iii) Identify the parts labelled **Q**, **R** and **S.**

(c) State **three** causes of accidents in the workshop
(d) State **one** reason for sprinkling water on sandcrete **blocks**.

[25 marks]

END OF ESSAY TEST

June 2016

BASIC DESIGN AND TECHNOLOGY 2
(Pre-Technical Skills 2)
Objective Test
ANSWERS

SECTION A
(CORE SKILLS)
[20 marks]
Answer question 1 [COMPULSORY]

1. (a) List four stitches that can be used for neatening as well as embroidery work. [4 marks]

- Loop / blanket stitches
- Herringbone stitches
- Buttonhole stitches
- Machine stitches
- Back stitches
- Overcasting
- Crotcheting

(b) List four dishes which are easily contaminated [2 marks]

- Sauces
- Gravies
- Bread pudding
- Vegetable turnovers
- Soups
- Meat turnovers
- Fish turnovers
- Milk dishes
- Vegetable salad
- Fruits salad

NB: any other protein food accepted

(c) (i) List three sources where information can be obtained to solve design problems [3 marks]

- Books and magazines
- Existing structre
- videos / films
- Questionnaire
- Interview
- Existing artefacts

(ii) List two methods of drawing objects in pictorial views. **[2 marks]**

- Oblique drawing
- Isometric drawing
- Perspective drawing

(iii) List two causes of weaknesses in wooden chairs. **[2 marks]**

- Use of unseasoned timber / defective timber
- Poor joints
- Poor construction (eg, missing nail, poor gluing, broken tongue)
- Improper use of the chair
- Over usage / old age

(d) (i) State one use of each of the elements listed below in a design **[4 marks]**

1. Shape: Used to create composition
2. Colour: Used for drawing and painting for beautification / code / identification
3. Space: Used in drawing and creating composition
4. Texture: Used to show the smoothness or roughness of an artwork/ appearance

(ii) Explain what a sketch is. **[2 marks]**

A sketch is a drawing mostly done quickly showing only essential details.

A quick drawing of an object or idea

(iii) State one reason why it is necessary to use a sketch when producing an object **[1 marks]**

- For planning
- To avoid waste
- For selection
- It saves time spent on the work
- It gives an overview of the end product

SECTION B
[50 marks]

Answer **two** *questions* **only** *from this section*

All questions carry equal marks

2. Figure 1 shows the front elevation and plan of an object made from a mild steel sheet.

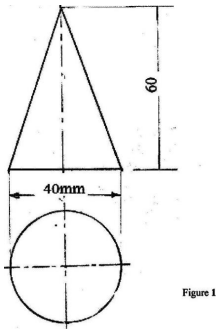


Figure 1

(d) Draw full size the following:

(i) front elevation

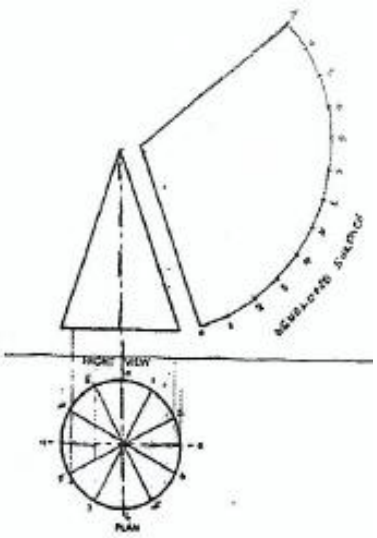
- drawing the base line
- drawing the two adjacent lines
- drawing the centre line

(ii) plan

- drawing the circle
- dividing the circle into equal number of parts

(iii) surface development

- drawing the seam
- drawing the arc
- stepping division from the plan onto the arc
- correct development of the cone



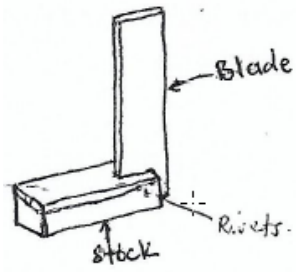
(e) (i) State the name of the object shown in Figure **1**

Cone

(ii) List **two** tools used for cutting the mild steel sheet.

- snips
- shears
- junior hacksaw
- sheet saw

(f) (i) Make a freehand pictorial sketch of a try square.



(ii) Indicate and label any two parts of the try square sketched in question 2(c)(i)

- Blade
- stock
- Rivets
- Brass plate

3. (a) List two tools each used in carrying out the following operations in blockwork:

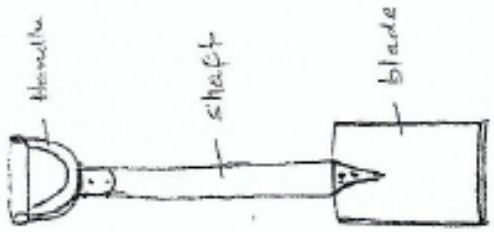
(i) laying

- hand trowel
- spirit level
- float
- gauge rod
- straight edge
- mortar board
- line and pins
- brick hammer / comb hammer
- bolster and club hammer
- builder's square

(ii) setting out.

- builder's square
- spirit level
- tape measure
- chalk
- pegs

(b) (i) Make a freehand pictorial sketch of a spade



(ii) Indicate and label any two parts of the spade sketched in question 3b(i)

- Handle
- Shaft
- Blade

(c) (i) State one use of the tenon saw.

- Used for cutting tenons
- For all general small cross cutting jobs

(ii) State one way of maintaining the tenon saw.

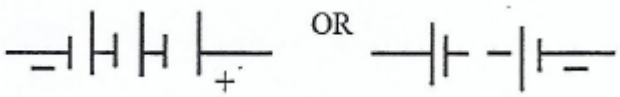
- Sharpening
- Shaping
- Setting
- Topping / jointing
- Oiling the blade
- Cleaning
- Racking the saw

(d) Sketch the representative symbol for each of the following electrical components:

(i) switch



(ii) battery.



(e) Copy and complete the table below:

Material	*One Property*	*One use*
(i) Clay	It swells when wet, it cakes with cracks when dried Shrinking ability is high. It does not adhere to cement. Soft, plastic and sticky when wet Shearing ability is high	Buildings, pots, local dishes, burnt bricks, roofing tiles, cement production, Blacksmith's forge
(ii) Wawa	Cream / Yellow-white in	Form work construction.

	colour. Interlocked and even-textured grain. Soft and easy to work with. It is not durable. Light in weight	Temporary structures in building construction sites. Inner veneers in plywood cheap furniture work.
(iii) Aluminium	Light in weight. Good conductor of electricity/ heat It is malleable. It is ductile. Corrosion resistant. Can be cast. Can be machined. Can be polished. Melts at 650°C. Shiny surface. High strength	Roofing sheet. Louvre frames. Louvre blades. Cooking utensils. Electrical cables. Cooking foil Parts of airplanes

4. (a) Copy and complete the table below by supplying one tool each for each of the following operations

Operation	*Blockwork*	*Woodwork*	*Metalwork*
(i) Cutting	Bolster, Brick/block hammer Wall chisel, Club hammer	Saws. Planes Chisels Gouges Brace bits	Hacksaws Files Chisels Shears Snips Drills Taps and dies
(ii) Measuring	Surveyors' tape Tape measure Gauge rod Builders square Gauge box Head pan	Rule Tape measure Folding rule Dividers Wing compasses Callipers Sliding bevel	Steel rule, Steel tape measure, Micrometer screw gauge Venier callipers Inside and outside callipers Pair of compasses
(iii) Striking	Club hammer Brick hammer	Claw hammer Warrington hammer Cross pein hammer Straight pein hammer Mallet	Ball pein hammer Sledge hammer Mallets

(b) Figure **2** shows the sketch of a handtool. *Use it to answer the following questions*:

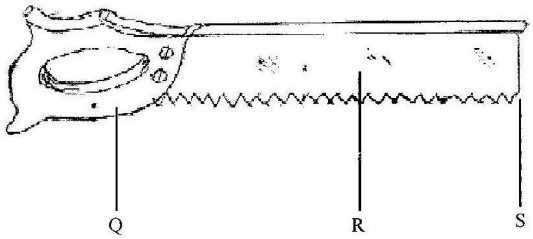


Figure 2

(i) Write the name of the tool.

Tenon saw

(ii) State one use of the tool

For cutting tenons,
For general small cross cutting joints

(iii) Identify the parts labelled Q, R and S.

Q - Handle
R - Blade
S - Toe

(c) State three causes of accidents in the workshop

- Slippery floors
- Running in the workshop
- Walking barefooted in the workshop
- Inattentiveness in the workshop
- Using wrong tools for the job
- Over confidence
- Not using protective clothing
- Using tools with faulty handles
- Carelessness

(d) State **one** reason for sprinkling water on sandcrete **blocks**.

- To help the block to attain its maximum strength through the process of curing
- To prevent rapid drying
- To reduce shrinkage
- To make it dense
- To prevent / reduce cracks
- To make it durable

June 2015

BASIC DESIGN AND TECHNOLOGY 1
(Pre-Technical Skills 1)

Objective Test
40 minutes

1. An example of a permanent stitch is
A. backstitch
B. basting
C. even tacking
D. tailors' tacks

2. The purpose of the napkin in table laying is to clean
A. sweat off the face
B. the mouth during eating
C. the plates before serving
D. the table after eating

3. Fish caught with chemicals may cause
A. diarrhoea
B. dysentery
C. poisoning
D. vomiting

4. Colours which are opposite to each other on the colour wheel are referred to as
A. analogous colours
B. complementary colours
C. cool colours
D. warm colours

5. An artist who has blue, red and yellow colours can obtain violet by mixing
A. all the three colours
B. blue with green
C. red with blue
D. yellow with blue

Use the diagram in Figure **1** *to answer questions* **6** *and* **7.**

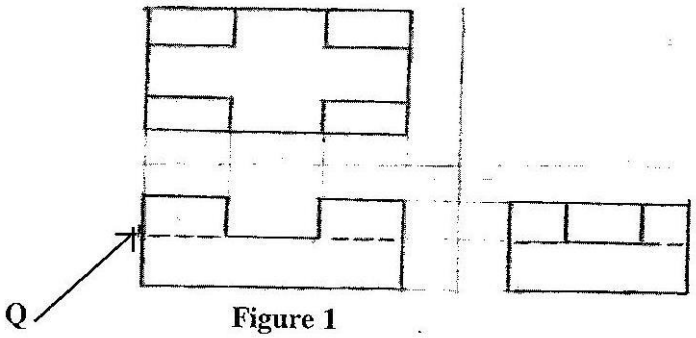


Figure 1

6. The type of projection used for the drawing is the
- A. first angle projection
- B. isometric projection
- C. oblique projection
- D. third angle projection

7. The view arrowed **Q** represents the
- A. front elevation
- B. plan
- C. left end view
- D. right end view

8. A sketch is a
- A. collection of different lines
- B. drawing made with pencil
- C. finished detailed drawing
- D. quick drawing in few lines

9. A developed cylinder shows a
- A. circle
- B. rectangle
- C. square
- D. triangle

10. Which of the following describes the strength and weakness of a design?
- A. Specification
- B. Investigation
- C. Evaluation
- D. Analysis

11. Shading is a rendering technique used to create
- A. outlines
- B. painting
- C. pictures
- D. tones

12. When a problem is described within any environment, it is termed as
- A. brief

B. situation
C. solution
D. specification

13. A place where artworks are displayed and sold is a
A. gallery
B. museum
C. studio
D. workshop

14. Texture, colour and shape in a design are referred to as
A. elements
B. ideas
C. objects
D. principles

15. The safest way of keeping one's daily sales is by
A. depositing it at the bank
B. giving it to mother for safe keeping
C. keeping it in an envelope
D. putting it under the sewing machine

16. An adhesive is used for
A. joining surfaces together
B. scaling against moisture
C. providing safety against fire hazard
D. providing surface decoration

17. Which of the following is a ferrous metal?
A. Brass
B. Bronze
C. Lead
D. Steel

18. The most suitable timber for decorative works is
A. odum
B. ofram
C. mahogany
D. wawa

Use the sketch in Figure **2** to answer questions **19** and **20**

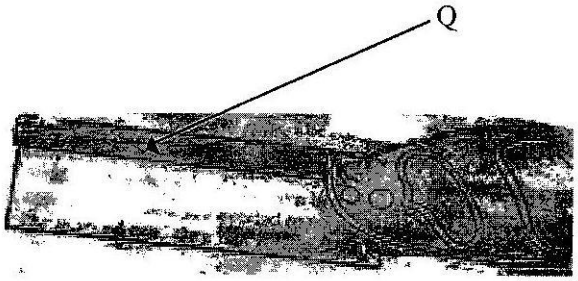


Figure 2

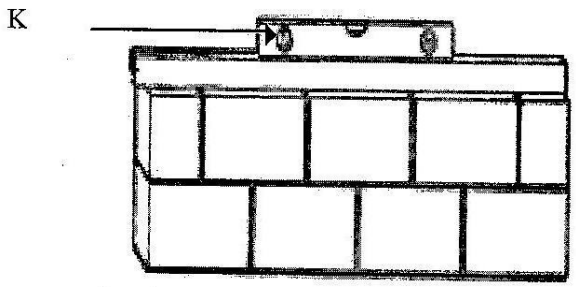
19. The tool is used for cutting
A. across thick boards
B. along thick boards
C. shoulders of tenons
D. thick soft metals

20. The purpose of the part labelled Q is to
A. control the blade
B. stiffen the blade
C. increase the width of the blade
D. protect the blade

21. The tool used to remove nail from wood is
A. brace bit
B. crowbar
C. pliers
D. screw driver

22. Sprinkling water on sandcrete blocks after moulding helps to
A. increase strength of blocks
B. increase weight of blocks
C. decrease weight of blocks
D. make bonding of blocks easier

Use figure **3** to answer questions **23** and **24**



23. Figure 3 shows the process of
A. checking the height of a wall
B. levelling a wall
C. plumbing a wall
D. setting out a wall

24. The tool labelled K is called
A. gauge rod
B. tamping rod
C. straight edge
D. spirit level

25. Soldering iron is made of copper because it is
A. a good conductor of heat
B. a good insulator
C. attractive to use
D. light to handle

26. A fuse wire breaks when there is
A. excess current
B. low current
C. low voltage
D. more resistance

27. In orthographic projection, the two principal planes intersect at
A. 30°
B. 45°
C. 60°
D. 90°

28. Pictorial drawings show objects in
A. one dimension
B. two dimension
C. three dimension
D. four dimension
E.

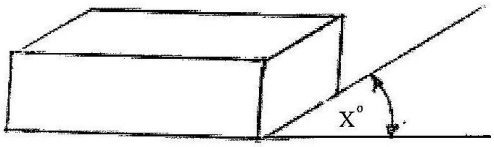


Figure 4

29. The angle x° shown on the oblique block sketched in Figure 4 is
A. 35°
B. 45°
C. 55°
D. 65°

30. Which of the following is the practical example of a cone?
A. Barrel
B. Funnel
C. Milk tin
D. Milo tin

June 2015

BASIC DESIGN AND TECHNOLOGY 1
(Pre-Technical Skills 1)

Objective Test

ANSWERS

1.	A.	backstitch
2.	B.	the mouth during eating
3.	C.	poisoning
4.	B.	complementary colours
5.	C.	red with blue
6.	D.	third angle projection
7.	A.	front elevation
8.	D.	quick drawing in few lines
9.	B.	rectangle
10.	C.	Evaluation
11.	D.	tones
12.	B.	situation
13.	A.	gallery
14.	A.	elements
15.	A.	depositing it at the bank
16.	A.	joining surfaces together
17.	D.	Steel
18.	B.	ofram
19.	C.	shoulders of tenons
20.	B.	stiffen the blade
21.	B.	crowbar
22.	A.	increase strength of blocks
23.	B.	levelling a wall
24.	D.	spirit level
25.	A.	a good conductor of heat
26.	A.	excess current
27.	D.	90°
28.	C.	three dimension
29.	B.	45°
30.	B.	Funnel

June 2015

BASIC DESIGN AND TECHNOLOGY 2
(Pre-Technical Skills 2)

Essay
1 hour 15 minutes

SECTION A
(CORE SKILLS)
[20 marks]
Answer question 1 [COMPULSORY]

1. (a) State **one** reason why a dressmaker would use a double stitch for fastening on a stitch instead of a knot.

[2 marks]

(b) List **two** suitable **each** methods of preserving the following food items:
(i) plantain
(ii) mango

[4 marks]

(c) (i) Copy and complete the table below by providing **one** suitable instrument for carrying out **each** of the operations listed:

OPERATION	ONE SUITABLE INSTRUMENT
Drawing a horizontal line	
Drawing a circle	
Measuring an angle	

(ii) Sketch the following geometrical solids:
(α) cylinder
(β) cone

[7 marks]

(d) (i) What is an exhibition? [2 marks]

(ii) Give **two** reasons why artworks are exhibited [2 marks]

(iii) List **two** types of exhibition [2 marks]

(iv) State **one** major difference between the two types of exhibition in (d)(iii) above.
[1 marks]

SECTION B
[50 marks]

Answer **two** *questions* **only** *from this section*

All questions carry equal marks

2. (a) Figure 1 shows the isometric view of a wooden stand.

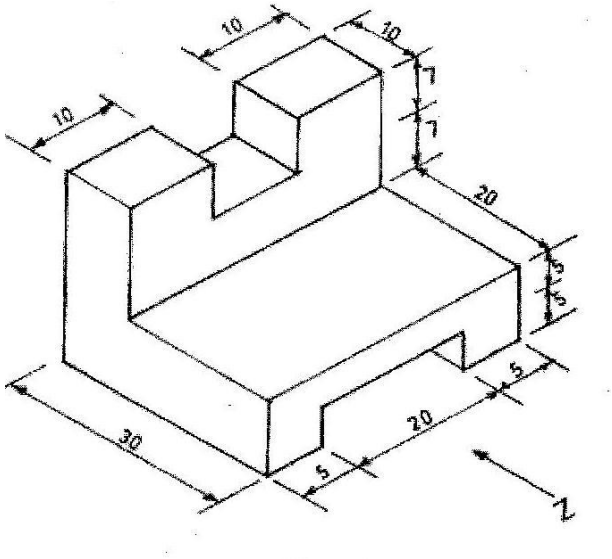


Figure 1

(i) Draw full size the front view in the direction of arrow Z

(ii) State the most suitable abrasive that can be applied on the work.

(b) State **one** method of preventing pinning in files

(c) (i) Make a freehand pictorial sketch of the wooden float.

(ii) Label any **two** parts of the tool sketched in (c)(i)

(iii) State **one** use of the wooden float

(d) (i) State **one** way of preventing accidents in the workshop

(ii) State **two** qualities of a good mortar

(iii) Explain the ration 1:6 in a mortar mix

3. (a) (i) Make a freehand pictorial sketch of a flat chisel

(ii) State **one** use of the flat chisel

(iii) State **one** method of maintaining the flat chisel

(b) Figure 2 shows the sketch of a type of sheet metal joint

(i) State the name of the joint

(ii) List **one** tool for cutting the sheet metal

(iii) State **one** article that can be made with the joint

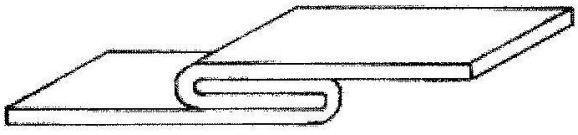


Figure 2

(c) Copy and complete the table below

	Name of tool	State one part of the tool	One use of the tool
(i)	Hacksaw		
(ii)	Soldering iron		
(iii)	Mortise chisel		

(d) State **one** reason each for the following operations
(i) glasspapering a piece of work made from wood;
(ii) mixing turpentine with oil paint [25 marks]

4. (a) (i) Make a freehand pictorial sketch of a marking knife
(ii) State **one** use of the marking knife
(iii) State **one** method of maintaining the marking knife.

(b) Copy and complete the table below

	Name of material	**One property each**	**One use each**
(i)	Wawa		
(ii)	Polyvinyl Chloride (PVC)		
(iii)	Clay		

(c) Figure 3 shows a sketch of a sandcrete block

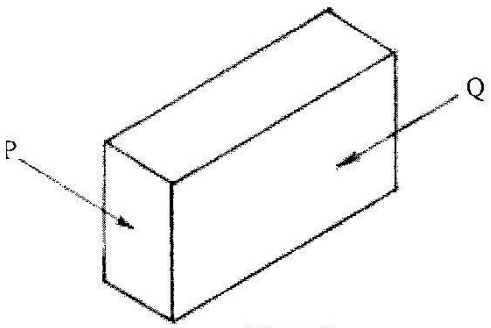


Figure 3

(i) State the parts labelled **P** and **Q**
(ii) List **two** materials used for making the block
(iii) Re-arrange the following procedures for moulding the block in their correct order:
1. tamp and use a wooden batten to strip off the excess mortar
2. batch sand and cement and mix thoroughly
3. remove and dry blocks
4. fill the mould box with mortar
5. add water and mix mortar;
6. assemble the mould box and clean the internal surface

[25 marks]

June 2015

BASIC DESIGN AND TECHNOLOGY 2
(Pre-Technical Skills 2)
Objective Test
ANSWERS

SECTION A
(CORE SKILLS)

1. (a) State one reason why a dressmaker would use a double stitch for fastening on a stitch instead of a knot. [2 marks]

- knots may come off from fabric
- knots may tear fine fabrics
- knots may cause tiny lumps in seams or hems when the fabric is pressed
- Double stitches are firmer / stronger so that it does not come off easily

(b) List two suitable each methods of preserving the following food items: [4 marks]

(i) plantain - Drying, freezing

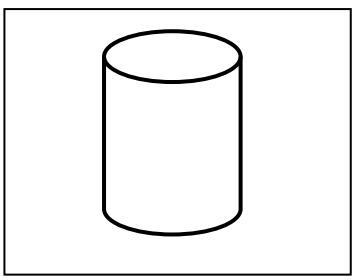
(ii) mango - Bottling, jam making, canning, drying chutney, freezing

(c) (i) Copy and complete the table below by providing one suitable instrument for carrying out each of the operations listed: [7 marks]

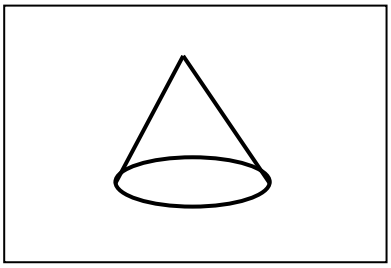
OPERATION	ONE SUITABLE INSTRUMENT
Drawing a horizontal line	Tee square (or T-square)
Drawing a circle	Pair of compasses
Measuring an angle	Protractor

(ii) Sketch the following geometrical solids: [7 marks]

(α) cylinder



(β) cone



(d) (i) What is an exhibition? [2 marks]

Public display of goods and services

(ii) Give two reasons why artworks are exhibited [2 marks]

Promote product or service
Advertise product or service
Sell product or service
Demonstrate product or service
Provide knowledge and skills about product or service

(iii) List two types of exhibition [2 marks]

- General
- Specialized

(iv) State one major difference between the two types of exhibition in (d)(iii) above. [1 mark]

- General exhibition deals with all kinds of products while
- Specialized deals with a particular product

SECTION B
[50 marks]

Answer **two** *questions* **only** *from this section*
All questions carry equal marks

2. (a) Figure 1 shows the isometric view of a wooden stand.

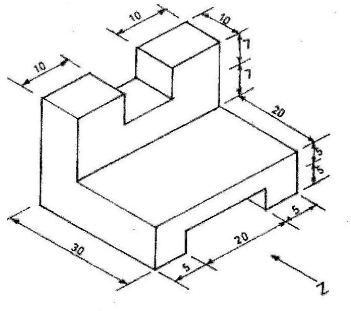
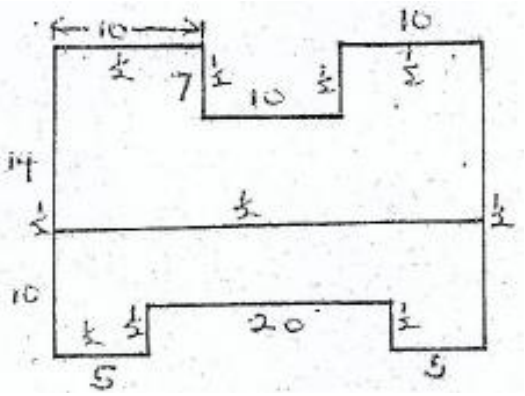


Figure 1

(i) Draw full size the front view in the direction of arrow Z

Drawing 7 horizontal lines and 6 vertical lines neatly



(ii) State the most suitable abrasive that can be applied on the work.

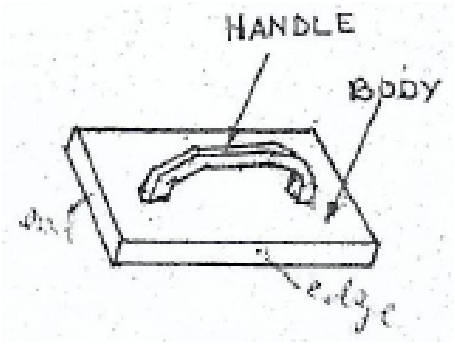
Glass paper
Sandpaper

(b) State one method of preventing pinning in files

Rubbing chalk on the file

(c) (i) Make a freehand pictorial sketch of the wooden float.

(ii) Label any two parts of the tool sketched in (c)(i)



(iii) State one use of the wooden float

- For rendering
- For plastering
- For filling mortar joints
- For holding mortar
- For screeding
- For dressing
- For spreading

(d) (i) State one way of preventing accidents in the workshop

- Wear protective clothing
- Avoid horse play
- Use right tool for the right job
- Avoid spilling oil on the floor

(ii) State two qualities of a good mortar

- workable
- free from impurities
- plastic / flexible
- uniform colour
- correct quantity of binding agents
- set within stipulated time

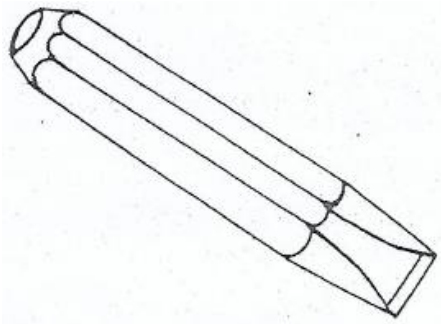
(iii) Explain the ration 1:6 in a mortar mix

It means one part of cement to six parts of sand.

or

Six parts of sand to one part of cement

3. (a) (i) Make a freehand pictorial sketch of a flat chisel



(ii) State one use of the flat chisel

- For cutting sheet metals
- For general purpose chiselling (metalwork / blockwork / woodwork)

(iii) State one method of maintaining the flat chisel

- Grind all burrs / mushroom heads
- Keep the cutting edge sharp
- Use it for the right job
- Keep it in a tool box or rack
- Clean with rag

(b) Figure 2 shows the sketch of a type of sheet metal joint



Figure 2

(i) State the name of the joint

Grooved lap joint

(ii) List one tool for cutting the sheet metal

- Shears / snips
- Flat chisel
- Sheet saw
- Junior hacksaw

(iii) State one article that can be made with the joint

- Milo tin
- Bucket
- Milk tin
- Metal trunk
- Dustbin
- Any food can
- Funnel

(c) Copy and complete the table below

	Name of tool	State one part of the tool	One use of the tool
(i)	Hacksaw	(i) Blade (ii) Frame (iii) Handle (iv)	(i) For cutting metals (ii) For cutting objects
(ii)	Soldering iron	(i) Handle (ii) Bit / head	For soldering

		(iii) Stem / shank	
(iii)	Mortise chisel	(i) Handle (ii) Cutting edge (iii) Blade (iv) Ferrule (v) Tang (vi) Washer	For cutting deep holes in Wood / mortising For chopping out waste in wood

(d) State one reason each for the following operations

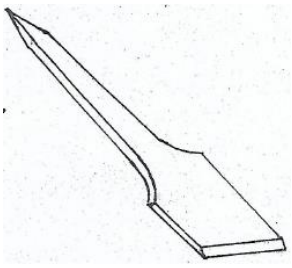
(i) glasspapering a piece of work made from wood;

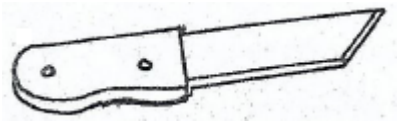
- To make it smooth
- To obtain good finish
- To make it attractive

(ii) mixing turpentine with oil paint

- To obtain uniform mixture
- To enable it flow easily under the brush
- For thinning oil paint / To dilute the paint

4. (a) (i) Make a freehand pictorial sketch of a marking knife





(ii) State one use of the marking knife

- It is used to mark lines on wood
- It is used to cut veneer

(iii) State one method of maintaining the marking knife.

- keep the cutting edge sharp
- clean after use
- keep it in a tool box
- oil the metal part

(b) Copy and complete the table below

	Name of material	One property each	One use each
(i)	Wawa	(i) Light in weight (ii) It has straight grains (iii) Easy to work with (iv) Less durable (v) White to pale yellow / cream in colour	(i) For making construction works (ii) For making Kitchen (iii) For making plywood (core)

(ii)	Polyvinyl Chloride (PVC)	(i) It is soft / flexible (ii) Easy to cut	(i) Making garden hose (ii) Electrical insulators (iii) Making suitcase (iv) Bowls
(iii)	Clay	(i) Smooth (ii) Sticky	(i) Traditional oven (ii) Traditional building (atakpame) (iii) Pottery work

(c) Figure 3 shows a sketch of a sandcrete block

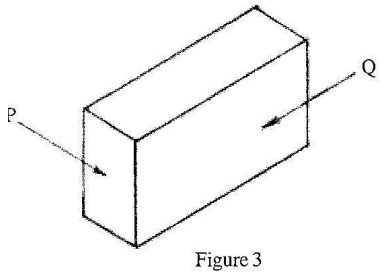


Figure 3

(i) State the parts labelled P and Q

P - header / header face
Q - stretcher / stretcher face

(ii) List two materials used for making the block

- sand
- cement
- water

(iii) Re-arrange the following procedures for moulding the block in their correct order:

1. **tamp and use a wooden batten to strip off the excess mortar**
2. **batch sand and cement and mix thoroughly**
3. **remove and dry blocks**
4. **fill the mould box with mortar**
5. **add water and mix mortar;**
6. **assemble the mould box and clean the internal surface**

1. batch sand and cement and mix thoroughly
2. add water and mix mortar
3. assemble the mould box and clean the internal surface
4. fill the mould box with mortar
5. tamp and use a wooden batten to strip off the excess mortar
6. remove and dry blocks

OR

1. - 2
2. - 5
3. - 6
4. - 4
5. - 1
6. - 3

June 2014

BASIC DESIGN AND TECHNOLOGY 1
(Pre-Technical Skills 1)

Objective Test
40 minutes

1. To hold a lining onto a fabric temporarily, one can use
- A. basting
- B. even tacking
- C. tailor's tacks
- D. thread marking

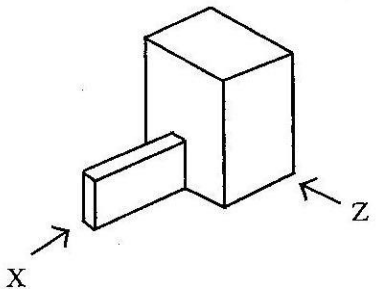
2. Which of the following stitches can be used to neaten raw edges of a set-in-sleeve?
- A. Feather
- B. Fishbone
- C. Herringbone
- D. Loop

3. The goiter on Evelyn's neck is caused by lack of
- A. calcium
- B. iodine
- C. protein
- D. vitamin

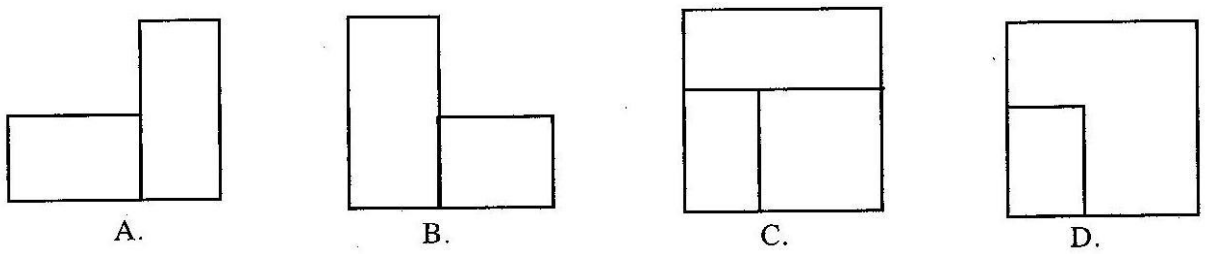
4. Man-made elements represented in art include
- A. branches
- B. rivers
- C. stones
- D. walls

5. The visitors' book signed at exhibitions helps the exhibitions helps the exhibitor to know the
- A. dignitaries who attended
- B. number of works in high demand
- C. observations made about the work
- D. people who came

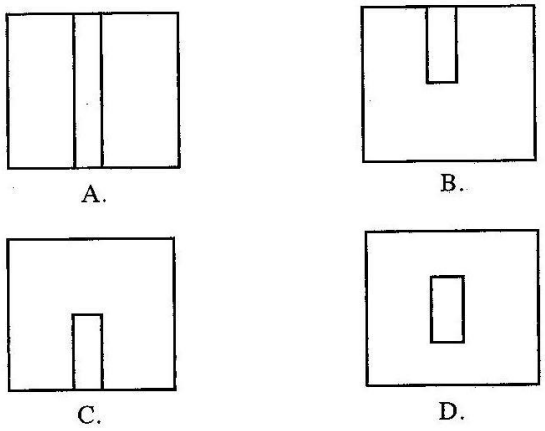
Use the sketch below to answer questions **6** *and* **7**



6. Which of the following represents the front view in the direction of arrow **Z**?



7. Which of the following represents the left end elevation in the direction of arrow **X**?



8. A place where art works are sold is

A. an archive
B. a gallery
C. a museum
D. a studio

9. In design, the conditions that the final solution must satisfy is termed

A. investigation
B. specification
C. development
D. analysis

10. Which of the following instruments is **most** suitable for drawing horizontal lines?

A. Tee-square
B. Protractor
C. Compass
D. Set-square

Figure 2 shows a design process. Use it to answer questions **11** *and* **12**

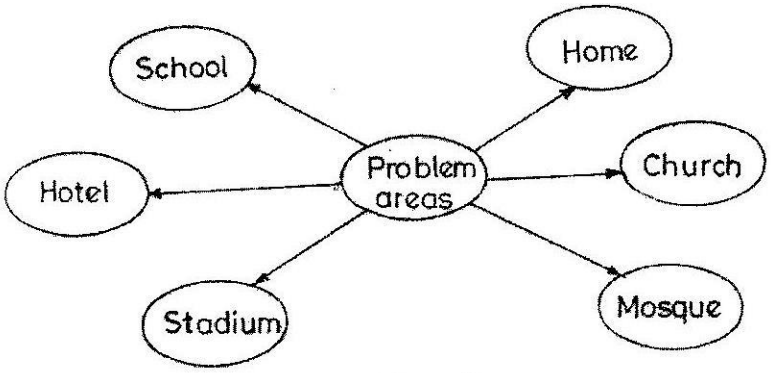


Figure 2

11. Figure 2 shows the method of
- A. specifying solutions
- B. stating specification
- C. evaluating artifact
- D. identifying needs

12. Which of the following design process stages follows the method shown in figure 2?
- A. Writing specifications
- B. Writing situation
- C. Analyzing a situation
- D. Sketching possible solutions

13. Shading with lines is described as
- A. cross hatching
- B. hatching
- C. stippling
- D. tonal value

14. Which of the following colours produces a feeling of warmth?
- A. Blue
- B. Green
- C. Violet
- D. White

15. Which of the following factors can cause a sewing business to collapse?
- A. Advertising
- B. Insurance
- C. Low sales
- D. Skilled labour

16. In mixing mortar, the ratio 1:5 represents
- A. one part sand to five parts cement
- B. one part water to five parts cement
- C. one part cement to five parts sand

D. one part cement to five parts gravel

17. Aggregates used for concrete mixture should be
A. well graded
B. round in size
C. square in size
D. medium graded

18. Which of the following represents a centre line?

A.

B.

C.

D.

19. Which of the following is a safety workshop practice?
A. Wearing strong boots
B. Wearing loose clothing
C. Leaving tools on bench
D. Using mushroom head chisel

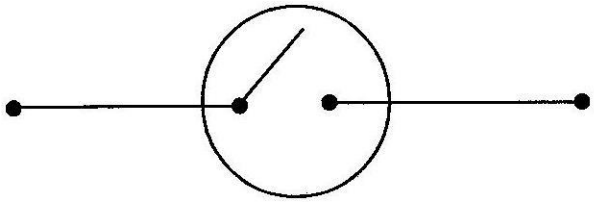
20. Polyvinyl acetate is also called
A. animal glue
B. white glue
C. resin glue
D. contact glue

21. Making a round hole in a piece of metal is termed
A. boring
B. chipping
C. drilling
D. chamfering

22. Which of the following is **not** a property of aluminium?
A. Light in weight
B. It corrodes
C. Absorbs heat readily
D. It is a good conductor of electricity

23. In furniture works, joints which have to be disconnected at certain times can be described as
A. permanent joints
B. immovable joints
C. movable joints
D. temporary joints

Use the symbol below to answer questions **24** *and* **25**



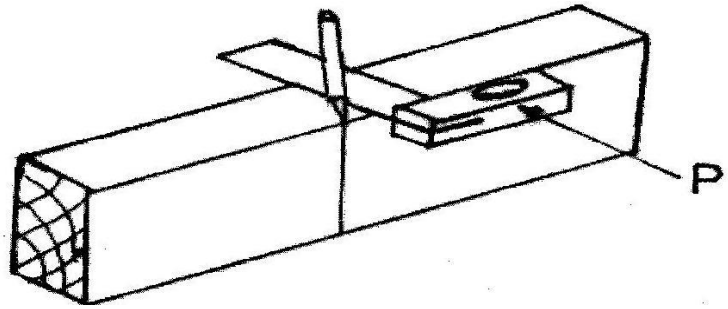
24. The symbol shows
- A. an inductor
- B. a diode
- C. a switch
- D. a capacitor

25. The component is used for
- A. storing electrical energy
- B. opening and closing a circuit
- C. opposing the flow of electric current.
- D. allowing current to flow in one direction

26. In applying finishes, the emulsion paint is mixed with
- A. oil
- B. water
- C. thinner
- D. turpentine

27. The **most** appropriate tool for enlarging holes in metals is the
- A. flat file
- B. hand file
- C. square file
- D. round file

Use the sketch below to answer questions **28** *to* **30**



28. The process represented in the sketch is
- A. marking out
- B. measuring
- C. cutting
- D. drilling

29. The tool marked P is a

A. marking gauge
B. builder's square
C. try-square
D. mortise gauge

30. The material being worked on is

A. polyvinyl chloride
B. mahogany
C. sandcrete block
D. mild steel.

END OF PAPER

June 2014

BASIC DESIGN AND TECHNOLOGY 1
(Pre-Technical Skills 1)
Objective Test
ANSWERS

1.	B.	even tacking
2.	C.	Herringbone
3.	B.	iodine
4.	D.	walls
5.	C.	observations made about the work
6.	A.	
7.	C.	
8.	B.	gallery
9.	B.	specification
10.	A.	Tee-square
11.	D.	identifying needs
12.	B.	Writing situation
13.	B.	hatching
14.	C.	Violet
15.	C.	Low sales
16.	C.	one part cement to five parts sand
17.	**A.**	well graded
18.	**A.**	
19.	**A.**	Wearing strong boots
20.	**B.**	white glue
21.	**C.**	drilling
22.	**B.**	It corrodes
23.	**C.**	movable joints
24.	**C.**	a switch
25.	**B.**	opening and closing a circuit
26.	**B.**	water
27.	**D.**	round file
28.	**A.**	marking out
29.	**C.**	try-square
30.	**B.**	mahogany

June 2014

BASIC DESIGN AND TECHNOLOGY 2
(Pre-Technical Skills 2)

Essay
1 hour 15 minutes

SECTION A
(CORE SKILLS)
[20 marks]
Answer question 1 [COMPULSORY]

1. (a) State the **main** method for cooking the following foods:

(i) rice
(ii) cake

(b) State **four** suitable methods for preserving fish.

(c) (i) List the type of pencil **most** suitable for the following operations:

OPERATION	PENCIL USED
Lettering	
Drawing outlines	
Shading	

(ii) List **four** methods of recording data for solving design problems.

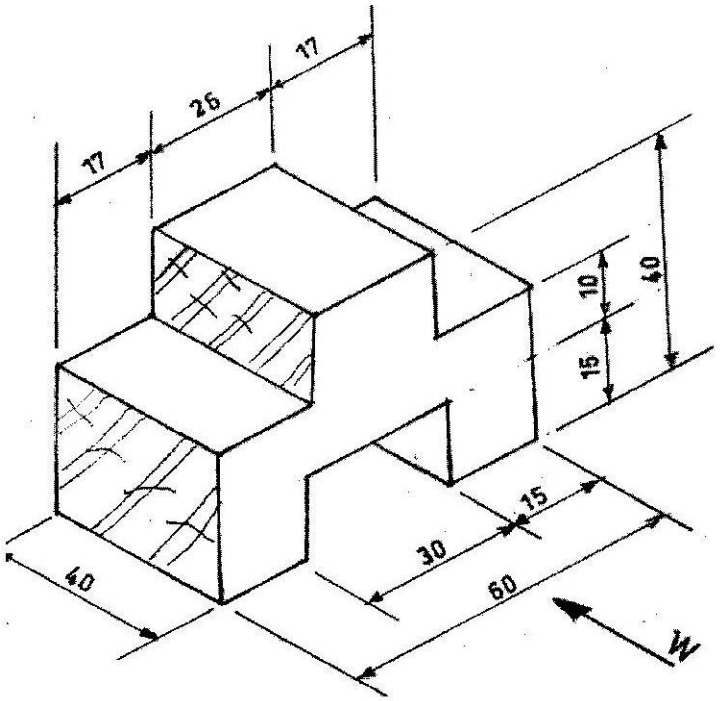
(d) (i) State **three** benefits of packaging.
(ii) List **two** important information on a good package

(e) Draw a line to illustrate **each** of the following:
(i) movement
(ii) texture.

SECTION B
[50 marks]

Answer **two** *questions* **only** *from this section*
All questions carry equal marks

2. *Figure* **1** *shows a wooden object to be prepared in the workshop.*



(a) Draw full size the front view in the direction of arrow **W**.

(b) List **two** tools used for:

(i) marking-out the shape of the object
(ii) cutting the waste wood.

(c) State **two** ways of caring for and maintaining the jack plane.
(d) State the difference between *ferrous metals* and *non-ferrous metals*.
(e) Make a freehand sketch of the builder's square.

3. (a) State **one** reason why personal safety should be observed in the workshop.

(b) List **two** specific tools for carrying out **each** of the following operations:
(i) setting-out a wall;
(ii) moulding bricks;
(iii) cutting a tenon from a piece of wood;
(iv) soldering sheet metals.

(c) (i) Make a freehand pictorial sketch of a rasp file

(ii) Label any two parts of the tool sketched in (c)(i) above.
(iii) State **one** use of the rasp file.

(d) (i) State the **main** difference between a brick and a block.
(ii) List the two **main** materials for making sandcrete blocks.

4. (a) List **one** protective clothing **each** for the following:
(i) foot;
(ii) head;
(iii) eyes,

(b) (i) Make a freehand pictorial sketch of the firmer chisel.
(ii) Label any **two** parts of the tool sketched in 4 (b) (i)

(c) State **one** reason **each** for carrying out the following operations:
(i) cross-filing
(ii) curing a sandcrete block;
(iii) setting out a wall.

(d) Copy and complete the table below

ALLOY	COMPOSITION	ONE USE EACH
Soft solder		
Brass		
Bronze		

June 2014

BASIC DESIGN AND TECHNOLOGY 2
(Pre-Technical Skills 2)
Objective Test
ANSWERS

SECTION A
(CORE SKILLS)

1. (a) State the **main** method for cooking the following foods:

(i) rice - boiling

(ii) cake - baking

(b) State **four** suitable methods for preserving fish.

- Drying
- Smoking
- Freezing
- Salting
- Frying
- Canning

(c) (i) List the type of pencil **most** suitable for the following operations:

OPERATION	PENCIL USED
Lettering	HB
Drawing outlines	H, 2H
Shading	2B or BB , 3B, 4B, 5B, 6B

(ii) List **four** methods of recording data for solving design problems.

- Taking notes / Using computer
- Taking photographs
- Video recording
- Audio recording
- Drawing / sketching
- Drawing graphs
- Making charts

(d) (i) State **three** benefits of packaging.

- It adds value to the product
- It protects the product from contamination
- It protects the consumer from hazards
- Attracts consumers
- Makes handling easier / makes it portable
- Promotes easy identification.

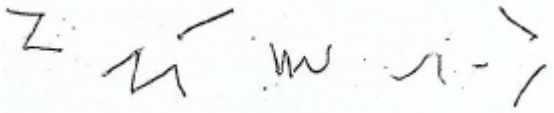
(ii) List **two** important information on a good package

- Information about content / manufacturer
- Date of production
- Expiry date
- Instructions for usage

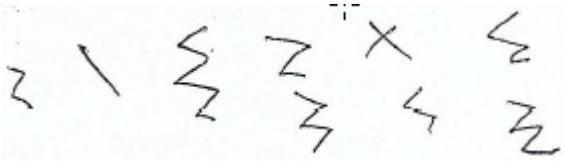
(e) Draw a line to illustrate **each** of the following:

(i) movement

Any line apart from horizontal or vertical lines, eg



(ii) texture.

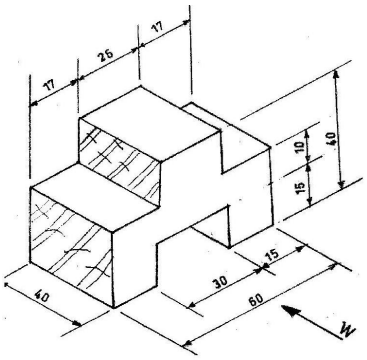


SECTION B
[50 marks]

Answer **two** *questions* **only** *from this section*

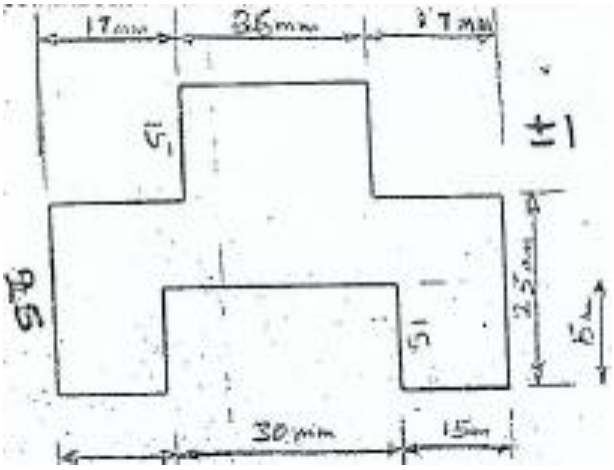
All questions carry equal marks

2. *Figure* **1** *shows a wooden object to be prepared in the workshop.*



(a) Draw full size the front view in the direction of arrow **W**.

Accurately drawing 6 horizontal lines and 6 vertical lines neatly



(b) List **two** tools used for:

(i) marking-out the shape of the object

- Marking knife
- Pencil
- Try-square
- Marking gauge
- Rule / Tape measure
- Straight edge

(ii) cutting the waste wood.

- Mortise chisel or firmer chisel
- Mallet
- Tenon saw / other back saws (except rip saw)

(c) State **two** ways of caring for and maintaining the jack plane.

- Cleaning the tool after use
- Oiling the metal parts
- Keeping the tool in the rack or tool box
- Sharpening the tool before use

(d) State the difference between *ferrous metals* and *non-ferrous metals*.

- Ferrous metals are metals that contain iron while non-ferrous metals do not contain iron.
- Ferrous metals can rust while non-ferrous metals do not rust.

(e) Make a freehand sketch of the builder's square.



3. (a) State **one** reason why personal safety should be observed in the workshop.

- To avoid injury
- To prevent accident

(b) List **two** specific tools for carrying out **each** of the following operations:

(i) setting-out a wall;

- Line and pins
- Straight edge
- Builder's square
- Profile Boards
- Spirit Level
- Tape measure / surveyor's tape
- Pegs
- Piece of chalk

(ii) moulding bricks;

- Mould box
- Head pan / gauge box / wheel barrow
- Shovel / spade
- Tamping rod
- Pick Axe

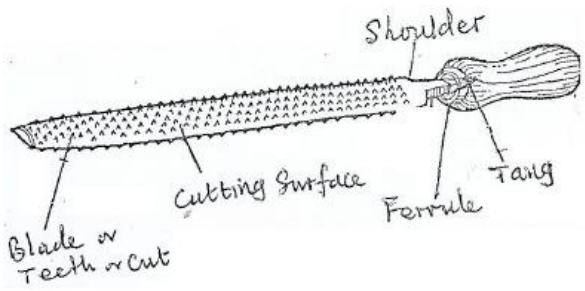
(iii) cutting a tenon from a piece of wood;

- Try square
- Marking gauge / mortise gauge
- Pencil / Marking knife
- Tenon Saw
- Mallet
- Bevel-edged chisel / firmer chisel

(iv) soldering sheet metals.

- Soldering iron / bit
- Coal pot / stove / blow lamp
- File / emery cloth
- Shears / pair of snips
- Mallet
- Folding bar

(c) (i) Make a freehand pictorial sketch of a rasp file



(ii) Label any two parts of the tool sketched in (c)(i) above.

- Blade / teeth / cut
- Tang
- Ferrule
- Handle
- Shoulder

(iii) State **one** use of the rasp file.

- For smoothening edges and shapes
- For shaping a circular piece of wood

(d) (i) State the **main** difference between a brick and a block.

- The size: A block is bigger than a brick
- The weight: A block is heavier than a brick
- The material: A block is made of concrete whilst brick is made of clay

(ii) List the two **main** materials for making sandcrete blocks.

- Sand
- Cement

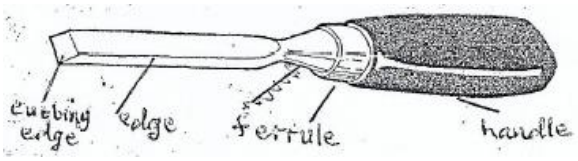
4. (a) List **one** protective clothing **each** for the following:

(i) foot; safety boots / safety shoes / Protective shoes

(ii) head; Helmet

(iii) eyes, Goggles / eye shield / spectacles

(b) (i) Make a freehand pictorial sketch of the firmer chisel.



(ii) Label any **two** parts of the tool sketched in 4 (b) (i)

- Cutting edge
- Edge
- Ferrule
- Handle

(c) State **one** reason **each** for carrying out the following operations:

(i) cross-filing

- For removing waste metal
- For removing unwanted metal
- For removing excess metal

(ii) curing a sandcrete block;

- To help the block to attain its maximum strength
- To prevent rapid drying
- To reduce shrinkage
- To make it dense
- To prevent / reduce cracks
- To make it durable

(iii) setting out a wall.

- For straightness
- For squaring
- For accuracy

(d) Copy and complete the table below

ALLOY	COMPOSITION	ONE USE EACH
Soft solder	Lead + tin or antimony	▪ For joining sheet metal ▪ For soldering sheet metal, wires cables and pipe
Brass	Copper + zinc	▪ Door locks ▪ Door handles ▪ Musical instruments ▪ Hinges ▪ Ornaments ▪ Taps ▪ Bolts and nuts ▪ Bells
Bronze	Copper + tin	▪ Statues ▪ Trophies ▪ Shields ▪ Church bells

June 2013

BASIC DESIGN AND TECHNOLOGY 1
(Pre-Technical Skills 1)

Objective Test
40 minutes

1. Which of the stitches can be used to hold down a hem decoratively?
- A. Basting
- B. Hemming
- C. Herringbone
- D. Overcasting

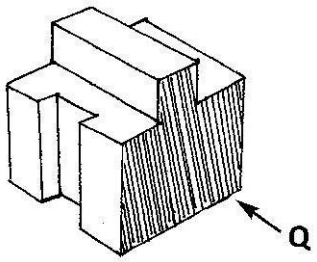
2. The advantage of buying food in bulk is that it
- A. becomes abundant
- B. becomes cheaper
- C. is easier to choose
- D. is easier to store

3. A balanced meal must contain
- A. food from each food group
- B. little fat and oil only
- C. protective foods only
- D. protein foods only

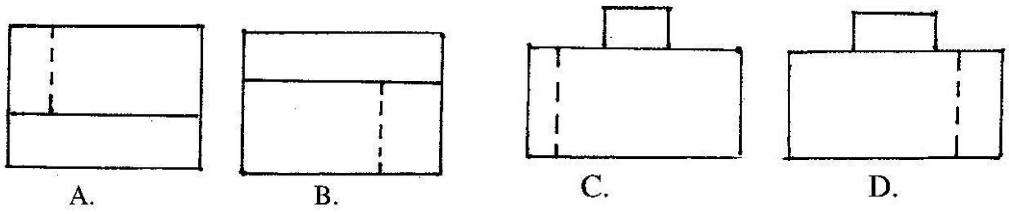
4. Drawing and painting are grouped under
- A. landscape
- B. still life
- C. three-dimensional work
- D. two-dimensional work

5. Identify the drawing tool from the following items
- A. Crayon
- B. Paper
- C. Palette
- D. Ruler

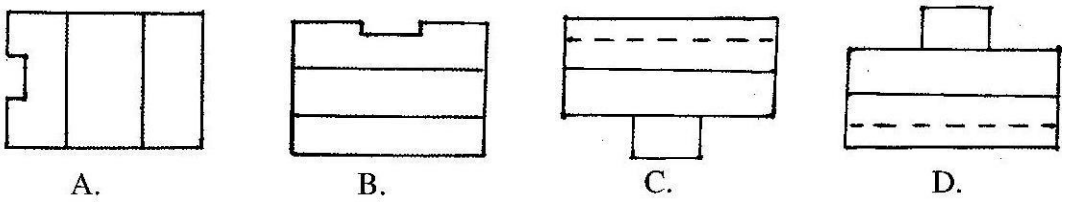
The figure below is a wooden block. Use it to answer Questions 6 and 7.



6. Which of the following represents the front view in the direction of arrow Q?



7. The plan is represented by



8. A sole proprietor

A. shares profit with friends
B. shares profit with a partner
C. takes decisions alone
D. takes decision with his staff

9. The statement describing the problem in designing is referred to as

A. design brief
B. design process
C. situation
D. investigation

10. Which of the following is **not** a type of joint for fixing parts of furniture ?

A. Movable joints
B. Unmovable joints
C. Temporary joints
D. Permanent joints

11. In oblique drawing one side of the object is inclined at an angle of

A. 30°
B. 45°
C. 60°
D. 90°

12. In a circuit diagram, current flows from

A. neutral to negative
B. neutral to positive
C. positive to negative
D. negative to positive

13. Select a neutral colour from the options below
A. blue
B. orange
C. red
D. white

14. The first step in designing is
A. choosing the right materials
B. colouring the sketches
C. identifying the problem
D. making sketch models

15. A dressmaker may also be skillful in
I. embroidery making
II. furnishing homes beautifully
III. making beautiful paintings
IV. selecting materials for upholstery

A. I, II and III only
B. I, II and IV only
C. I, III and IV
D. II, III and IV only

16. Turpentine is mixed with
A. emulsion paint
B. lacquer
C. oil paint
D. thinner

17. The elevation projected onto the side vertical plane is the
A. front
B. plan
C. end
D. back

18. Which of the following is not a fastener?
A. bolts and nuts
B. soldering bits
C. screws
D. rivets

19. Files are cleaned by using the
A. file card

B. hacksaw blade
C. scriber
D. chisel

20. Which of the following components stores electrical charge?
A. diode
B. switch
C. resistor
D. capacitor

21. The type of material used to strengthen joints in woodwork is known as
A. adhesives
B. abrasives
C. sanding sealer
D. wax polish

22. Which of the following types of line is used to indicate the size of a drawing?
A. Hidden line
B. Centre line
C. Construction line
D. Dimension line

23. Sandcrete bricks are made from
A. cement and clay
B. cement and sand
C. cement and laterite
D. cement and lime

24. Which of the following is used to protect the head from injury?
A. apron
B. gloves
C. goggles
D. helmet

25. The funnel is an example of surface development of a
A. cone
B. cylinder
C. square prism
D. square pyramid

26. Which of the following tools are used for boring wood?
A. brace and bit
B. mortice chisel and mallet
C. hammer and cold chisel
D. hammer and centre punch

27. The first step to be taken when checking a socket outlet for a fault is to
A. open the socket outlet

B. switch off power at the main switch
C. test the socket outlet
D. remove the main fuse

28. The soft solder is an alloy of
A. copper and lead
B. copper and tin
C. lead and tin
D. lead and aluminium

29. Which of the following is a cutting tool?
A. scrubbing plane
B. bench vice
C. rawhide mallet
D. soldering iron

30. A face side mark on planed wood indicates that the surface
A. is smooth
B. is rough
C. has been tested for flatness
D. has been tested for squareness

June 2013

BASIC DESIGN AND TECHNOLOGY 1
(Pre-Technical Skills 1)

Objective Test

ANSWERS

1.	C.	Herringbone
2.	B.	becomes cheaper
3.	A.	food from each food group
4.	D.	two-dimensional work
5.	A.	Crayon
6.	C.	
7.	A.	
8.	C.	takes decisions alone
9.	C.	situation
10.	C.	Temporary joints
11.	B.	45°
12.	C.	positive to negative
13.	D.	white
14.	C.	identifying the problem
15.	B.	I, II and IV only
16.	C.	oil paint
17.	C.	end
18.	B.	soldering bits
19.	A.	file card
20.	D.	capacitor
21.	A.	adhesives
22.	D.	Dimension line
23.	B.	cement and sand
24.	D.	helmet
25.	A.	cone
26.	A.	brace and bit
27.	B.	switch off power at the main switch.
28.	C.	lead and tin
29.	A.	scrubbing plane
30.	C.	has been tested for flatness

June 2013

BASIC DESIGN AND TECHNOLOGY 2
(Pre-Technical Skills 2)

Essay
1 hour 15 minutes

SECTION A
(CORE SKILLS)
[20 marks]
Answer question 1 [COMPULSORY]

1. (a) State two functions of protein in the body

(b) List two ways by which patching can be done

(c) (i) Explain a design brief
(ii) List three methods of carrying out design investigations.
(iii) State two factors to consider when selecting the best solution from the generated possible solutions in designing.

(d) (i) What is the colour spectrum?
(ii) List the three primary colours
(iii) Explain secondary colours

SECTION B
[50 marks]
Answer **two** *questions* **only** *from this section*
All questions carry equal marks

2. Figure 1 shows two views of a solid to be produced with sheet metal

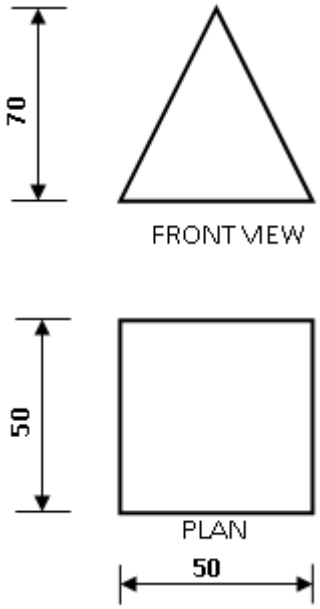


FIGURE 1

(a) Draw full size the following:
- (i) front view
- (ii) plan
- (iii) surface development

(b) State the name of the solid in figure 1

(c) (i) make a freehand pictorial sketch of a marking gauge
- (ii) Label any two parts of the tool you have sketched in (c)(i) above

3. (a) Using symbols, draw an electrical circuit diagram to show the following
- (i) two bulbs in parallel
- (ii) a switch
- (iii) a cell

(b) State one function of a light emitting diode (LED) in an electrical circuit.

(c) (i) Make a freehand sketch of a dot punch.

(ii) Indicate the point angle on the tool you sketched in (c)(i) above

(d) Copy and complete the table below

ITEM	ONE SUITABLE MATERIAL FOR MAKING IT
Traditional oven	
Kitchen stool	
Bristle of tooth brush	
Garden rake	

(e) State one reason each for carrying out the following processes
- (i) leveling a wall

(ii) applying flux when soldering

4. (a) Explain the following

(i) ratio 1:4 for mortar mixture;

(ii) non-ferrous alloys

(b) (i) Make a freehand pictorial sketch of a flat screwdriver

(ii) Label two parts of the tool you sketched in (b)(i) above

(iii) State one main use of the tool you sketched in (b)(i) above.

(c) (i) Add three more courses to the wall in Figure 2

(ii) Show a toothing end on the wall

(iii) List two tools for laying the wall

(iii) State one reason for intruding a half-bat in the wall.

June 2013

BASIC DESIGN AND TECHNOLOGY 2
(Pre-Technical Skills 2)

Essay Test

ANSWERS

SECTION A
(CORE SKILLS)

1. (a) State two functions of protein in the body

It promotes growth and development.
It repairs worn out tissues in the body.
It provides energy in the absence of carbohydrates
It supports production of hormones and enzymes

(b) List two ways by which patching can be done

By hand
By machine

(c) (i) Explain a design brief

A short statement that expresses the solution to a problem
Or
A statement of what the designer intends to design and make

(ii) List three methods of carrying out design investigations.

By observing carefully
By visiting
By experimenting
By reading relevant literature
By interviewing
By surfing the internet

(iii) State two factors to consider when selecting the best solution from the generated possible solutions in designing.

Its function / effectiveness
Its physical appearance (Aesthetics)
Its cost of designing and making
Its safety and suitability
Its durability

(d) (i) What is the colour spectrum?

The distribution of colours produced when white light is dispersed by a prism

(ii) List the three primary colours?

The primary colours for

(α) pigment: red, yellow and blue

(β) light (additive): red, green and blue

(γ) light (subtractive): magenta, cyan and yellow

(iii) Explain secondary colours

A colour resulting from the mixing of two primary colours in equal proportions

For pigment, the secondary colours are

Violet = (red + blue),

green = (yello + blue) and

orange = (red + yellow)

For light (additive), the secondary colours are

Yellow = (green + red)

Cyan = (green + blue)

Magenta = (red + blue)

For light (subtractive), the secondary colours are:

Red = (yellow + magenta)

Green = (yellow + cyan)

Blue = (magenta + cyan)

SECTION B
[50 marks]

Answer **two** *questions* **only** *from this section*

All questions carry equal marks

2. Figure 1 shows two views of a solid to be produced with sheet metal

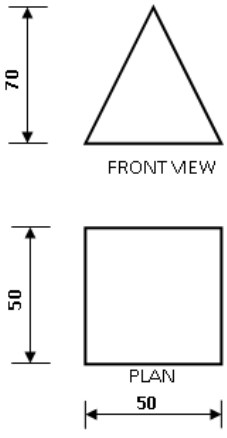


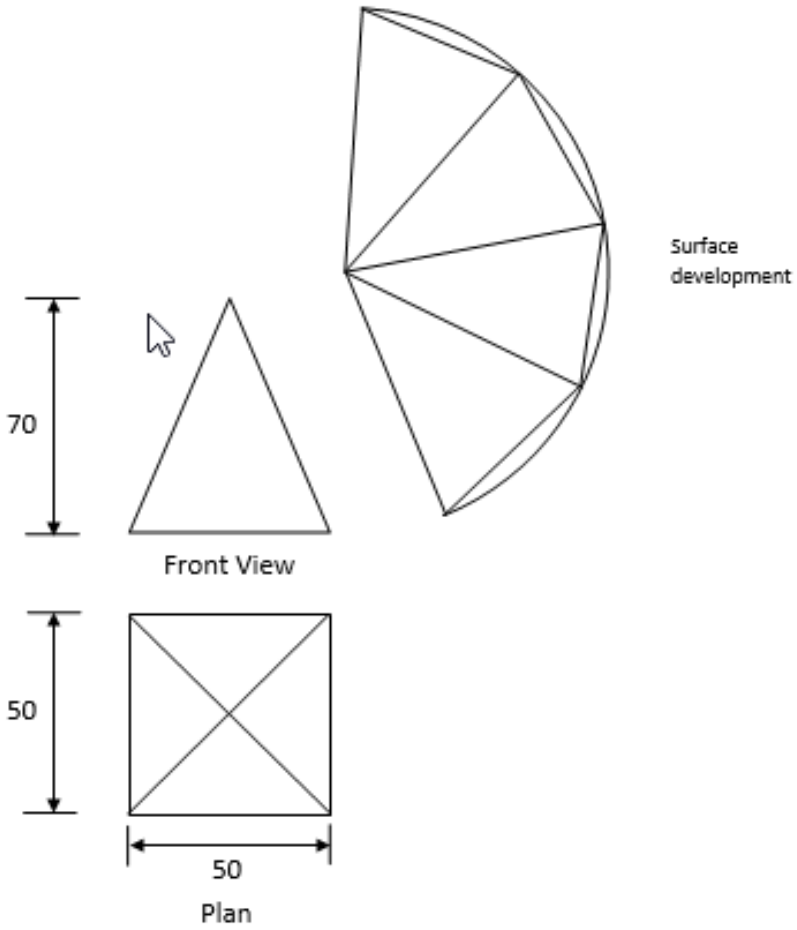
FIGURE 1

(a) Draw full size the following:

(i) front view

(ii) plan

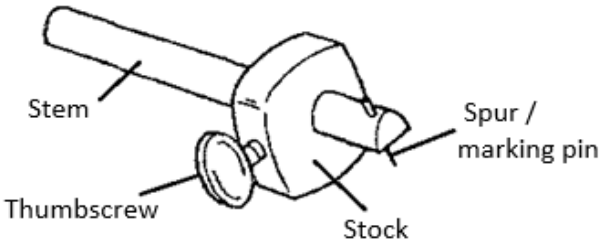
(iii) surface development



(b) State the name of the solid in figure 1

Square pyramid

(c) (i) make a freehand pictorial sketch of a marking gauge



(ii) Label any two parts of the tool you have sketched in (c)(i) above

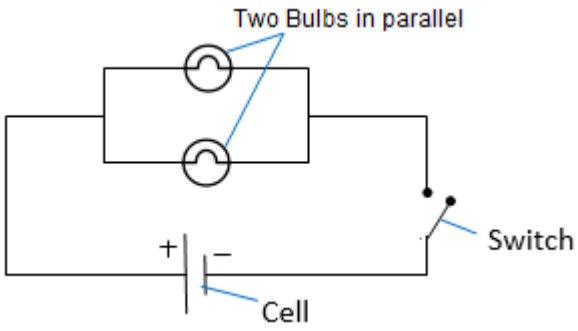
- Thumbscrew
- Stem
- Spur / marking pin
- Stock

3. (a) Using symbols, draw an electrical circuit diagram to show the following

(i) two bulbs in parallel

(ii) a switch

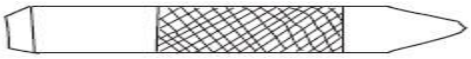
(iii) a cell



(b) State one function of a light emitting diode (LED) in an electrical circuit.

- Used to indicate the presence of electric current in the electrical circuit.
- It allows electric current to flow through in only one direction.

(c) (i) Make a freehand sketch of a dot punch.



(ii) Indicate the point angle on the tool you sketched in (c)(i) above

60°

(d) Copy and complete the table below

ITEM	ONE SUITABLE MATERIAL FOR MAKING IT
Traditional oven	Clay
Kitchen stool	Odum, wawa
Bristle of tooth brush	Nylon
Garden rake	Iron, cast iron, steel

(e) State one reason each for carrying out the following processes

(i) leveling a wall

To ensure that the wall is perfectly horizontal

OR

To check the horizontal straightness of the wall.

(ii) applying flux when soldering

- To prevent oxidation (the formation of fresh oxides) by forming a protective layer on the solder.
- To clean the surface of the base metal during heating.
- To eliminate impurities appearing on metal surface.
- To breakdown the surface tension of molten solder which helps the easy flow.

4. (a) Explain the following

(i) ratio 1:4 for mortar mixture;

One part of cement to be mixed with four parts of sand.

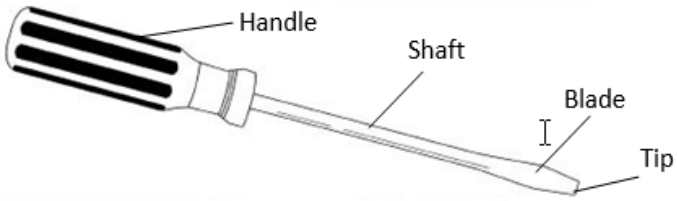
(ii) non-ferrous alloys

Non ferrous alloys are alloys that do not contain iron

OR

Non ferrous alloys are uniform mixtures of metals that do not contain iron

(b) (i) Make a freehand pictorial sketch of a flat screwdriver



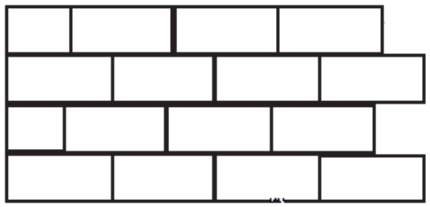
(ii) Label **two** parts of the tool you sketched in (b)(i) above

- Handle
- Shaft
- Blade
- Tip

(iii) State one main use of the tool you sketched in (b)(i) above.

It is used to drive in / tighten and drive out / loosen screws into/out of work peices

(c) (i) Add three more courses to the wall in Figure 2



(ii) Show a toothing end on the wall

(iii) List two tools for laying the wall

- hand trowel
- spirit level
- float
- gauge rod
- straight edge
- mortar board
- line and pins
- brick hammer / comb hammer
- bolster and club hammer
- builder's square

(iii) State one reason for intruding a half-bat in the wall.

To avoid continuous vertical joints

April 2012

BASIC DESIGN AND TECHNOLOGY 1
(Pre-Technical Skills 1)

Objective Test
40 minutes

1. Which of the sets are temporary stitches?
- A. Basting and tracking
- B. Hemming and basting
- C. Tracking and hemming
- D. Tracking and running

2. The main vitamin Ama gets when she eats oranges is
- A. vitamin C
- B. vitamin D
- C. vitamin E
- D. vitamin K

3. Crockery is a collection of
- A. forks and spoons
- B. glasses and spoons
- C. plates and forks
- D. plates and glasses

4. The arrangement of the shapes in a picture is known as:
- A. blending
- B. composition
- C. drawing
- D. toning

5. Which of the following colours produces a feeling of coolness?
- A. blue
- B. orange
- C. violet
- D. white

6. The aspect of analysis which considers the appearance of an artefact is known as
- A. aesthetics
- B. construction
- C. ergonomics
- D. function

7. In designing, possible solutions are generated by the use of

I. observation
II. drawing
III. sketches

Which of the following options are correct?

A. I and II only
B. II and III only
C. I and III only
D. I, II and III

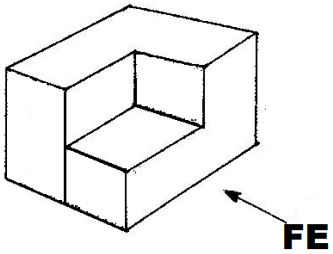
8. Drawing of an object starts with

A. colour
B. line
C. shape
D. texture

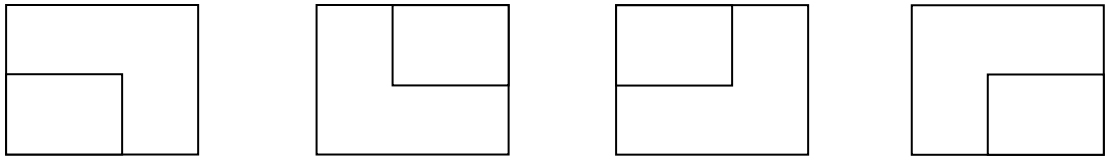
9. Which of the following colours is used to represent an earth wire?

A. blue
B. brown
C. red
D. yellow

The figure below is a wooden block. Use it to answer question 10 and 11

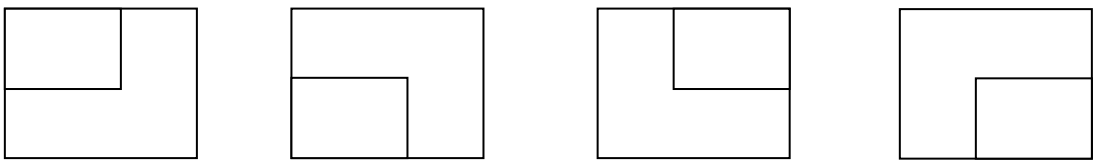


10. Which of the following views represent the plan?



A B C D

11. The left-end view is represented by



A B C D

12. Weakness on a joint of furniture may be due to

A. excessive glueing
B. hammering hard
C. missing nail
D. over polishing

13. Which of the following options suggest a dot in nature?

A. a cube of sugar
B. a grain of sand
C. footpath lane
D. Veins of a leaf

14. The B range of pencils are the

A. finest
B. hardest
C. smoothest
D. softest

15. Mounting an exhibition enables the dressmaker to

I. show her handiwork
II. attract customers
III. increase her income
IV. show her generosity

A. I, II and III only
B. I, II and IV only
C. II and IV only
D. III and IV only

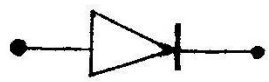
16. Brass is made of

A. copper and tin
B. copper and zinc
C. lead and tin
D. lead and zinc

17. Which of the following tools is used for checking alignment of a wall?

A. building square
B. folding rule
C. gauge
D. straight edge

18. The symbol below indicates a/an



A. diode
B. capacitor
C. inductor
D. resistor

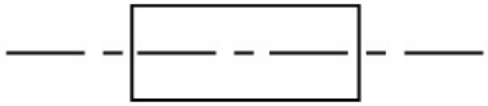
19. Iron filings are removed from the teeth of a file by using

A. chalk
B. scriber
C. file card
D. flat chisel

20. Mould boxes are oiled before use to

A. increase the size of the brick
B. prevent brick sticking to the box
C. smoothen the surface of the brick
D. give the brick a shiny surface

21. The type of line drawn across the rectangle below is



A. centre line
B. hidden detail
C. construction line
D. dimensional line

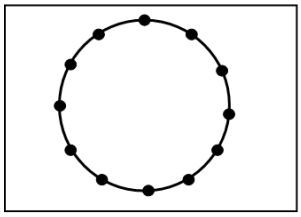
22. Which of the following is an abrasive?

A. oil paint
B. emery cloth
C. wax polish
D. contact glue

23. Metals which contain iron and carbon are called

A. non-ferrous alloy
B. non-ferrous metals
C. ferrous metals
D. fibre glass

24. The tool used for marking out the final circular shape below is a



A. scriber
B. pair of dividers
C. dot punch
D. centre punch

25. Glass paper is used on wood to

A. fill the pores
B. produce a smooth surface
C. form beautiful grains
D. reduce the thickness

26. Which of the following is **not** a safety measure when using electrical appliances?

A. Read all warning and instructions on the appliance before use
B. Do not touch electrical appliance with wet hands
C. Do not plug in too many appliances on a socket at a time
D. Leave all appliances on after use.

27. Which of the following is the correct order of applying a finish to a block wall?

I. Plastering
II. Painting
III. Scraping

A. I, II, III
B. I, III, II
C. II, I, III
D. III, II, I

Use the figure below to answer questions **28** *to* **30**



28. The figure is a

A. rule
B. tee square
C. try square
D. set square

29. The part labeled X is called

A. blade
B. lead
C. set
D. stock

30. The tool shown in the figure is used to

A. draw horizontal lines
B. draw vertical lines
C. set variable angles
D. measure horizontal lines

April 2012

BASIC DESIGN AND TECHNOLOGY 1
(Pre-Technical Skills 1)
Objective Test
ANSWERS

1.	A.	Basting and tacking
2.	A.	vitamin C
3.	D.	plates and glasses
4.	B.	composition
5.	A.	blue
6.	A.	aesthetics
7.	B.	II and III only
8.	B.	line
9.	D.	yellow
10.	A.	
11.	C.	
12.	C.	missing nail
13.	B.	a grain of sand
14.	D.	softest
15.	A.	I, II and III only
16.	B.	copper and zinc
17.	D.	straight edge
18.	A.	diode
19.	C.	file card
20.	B.	prevent brick sticking to the box
21.	A.	centre line
22.	B.	emery cloth
23.	C.	ferrous metals
24.	D.	centre punch
25.	B.	produce a smooth surface
26.	D.	Leave all appliances on after use
27.	B.	I, III, II
28.	B.	tee square
29.	D.	stock
30.	A.	draw horizontal lines

April 2012

BASIC DESIGN AND TECHNOLOGY 2
(Pre-Technical Skills 2)

Essay
1 hour 15 minutes

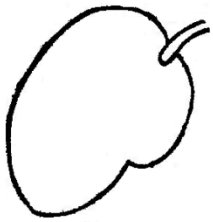
SECTION A
(CORE SKILLS)
[20 marks]
Answer question 1 [COMPULSORY]

1. (a) Ama ate fried fish, kenkey and fresh pepper sauce

Copy and complete in a table what Ama ate into the functional groups of food.

Food item	Functional group
(i) Fried fish	
(ii) Kenkey	
(iii) Fresh pepper sauce	

(b) The diagram below is the drawing of a mango



(i) Enlarge the diagram in (b) in the drawing sheet provided below.

(ii) Shade your drawing in (b) (i) in **three** tones

(iii) List **two** other methods of shading apart from the one used in (b) (ii)

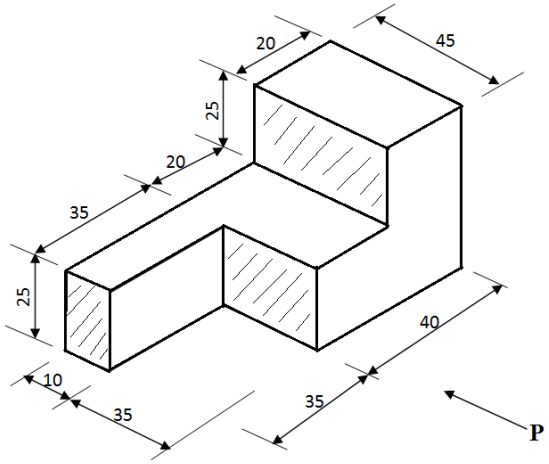
(c) (i) State **two** reasons why investigations are carried out in the design process.

(ii) List **three** electrical appliances used in the home

SECTION B
[50 marks]

Answer **two** *questions* **only** *from this section*
All questions carry equal marks

2. The figure shows the pictorial view of a wooden block made from odum



(a) (i) Draw full size the front elevation in the direction of arrow P

(ii) List **two** finishes that can be applied on the wooden block

(b) State **one** use each of the following in soft soldering:

(i) flux;
(ii) soldering bit

(c) Make a freehand sketch of a pair of snips

(d) The table below shows tools, their uses and ways of caring for them.

Copy and complete the table

Tool	One use each	One way of caring for the tool
(i) Builders square		
(ii) Hand file		
(iii) Rip saw		
(iv) Gauge rod		

3. (a) Explain **each** of the following terms as used in concrete :
- (i) fine aggregates
- (ii) coarse aggregates

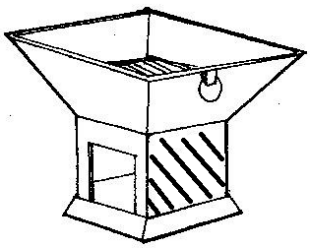
(b) List **three** methods of pictorial drawing

(c)
- (i) Make a freehand sketch of a pick-axe
- (ii) Label any **two** parts on the tool sketched in (c)(i) above

(d) List **two** items made of cast iron.

(e) Draw the symbol for each of the following electronic components
- (i) resistor
- (ii) capacitor
- (iii) a switch

(f) The figure below shows a pictorial view of a coal-pot made from a metal plate



Arrange the following steps in a sequential order for making the coal-pot.
- (i) File pieces to the required sizes;
- (ii) Mark-out the parts of the coal-pot
- (iii) Assemble the parts;
- (iv) Measure the metal pieces to the required dimensions;
- (v) Cut out pieces.

4. (a)
- (i) State the difference between plumbing and leveling in walling
- (ii) List **two** tools used for plumbing and leveling in walling

(b) List **two** examples each of a conductor and an insulator

(c) State **one** purpose each for the following

(i) curing a brick;

(ii) applying a piece of chalk on the face of a file.

(d) (i) Make a freehand sketch of the soldering iron used in sheet metal work.

(ii) Label any **two** parts on the tool sketched in (d) (i)

(e) The table below shows two materials to be glued together

Copy and complete the table

Prepared Materials	**Types of adhesive**
(i) Odum to odum	
(ii) Formica to wawa	
(iii) Veneer to odum	

April 2012

BASIC DESIGN AND TECHNOLOGY 2
(Pre-Technical Skills 2)

Essay

ANSWERS

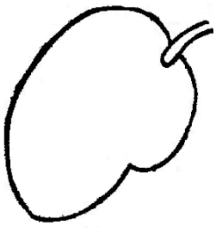
SECTION A
(CORE SKILLS)

1. (a) Ama ate fried fish, kenkey and fresh pepper sauce

Copy and complete in a table what Ama ate into the functional groups of food.

Food item	Functional group
(i) Fried fish	Body building food
(ii) Kenkey	Energy giving food
(iii) Fresh pepper sauce	Protective food

(b) The diagram below is the drawing of a mango



(i) Enlarge the diagram in (b) in the drawing sheet provided below.

(ii) Shade your drawing in (b) (i) in **three** tones



(iii) List **two** other methods of shading apart from the one used in (b) (ii)

Hatching (shading with lines)
Cross hatching (shading with crossed lines)
Stippling (shading with dots)
Smudging (shading by smearing with finger)

(c) (i) State **two** reasons why investigations are carried out in the design process.

- To properly identify and specify the problem
- To decide on best solutions to the problem
- To decide on the most suitable materials and tools needed to solve the problem
- To decide on the right form/shape and size of the article

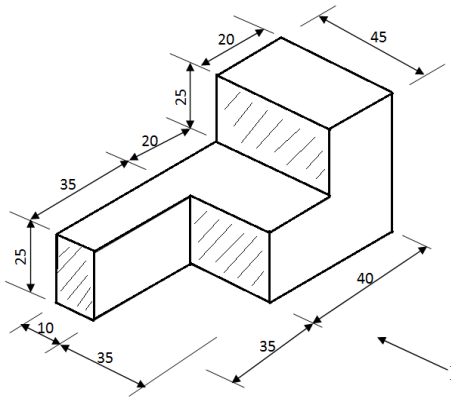
(ii) List **three** electrical appliances used in the home

- Pressing iron
- Radio set
- Television set
- Electric cooker
- Refrigerator
- Electric kettle
- Blender
- Microwave oven

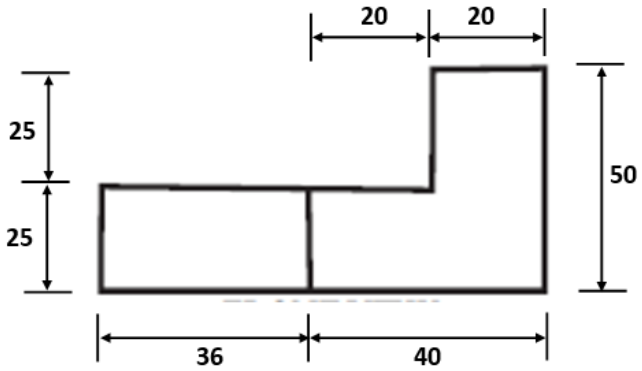
SECTION B
[50 marks]

Answer **two** *questions* **only** *from this section*
All questions carry equal marks

2. The figure shows the pictorial view of a wooden block made from odum



(a) (i) Draw full size the front elevation in the direction of arrow P



(ii) List **two** finishes that can be applied on the wooden block

- Varnish
- Wax
- Lacquer
- Oil paint

(b) State **one** use each of the following in soft soldering:

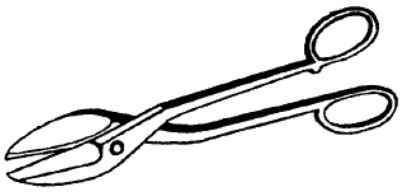
(i) flux;

- It prevents oxidation (the formation of fresh oxides) by forming a protective layer on the solder.
- It cleans the surface of the base metal during heating.
- It eliminates impurities appearing on metal surface.
- It breakdowns the surface tension of molten solder which helps the easy flow.

(ii) soldering bit

- It gives the necessary heat to melt the solder

(c) Make a freehand sketch of a pair of snips



OR



(d) The table below shows tools, their uses and ways of caring for them.

Copy and complete the table

Tool	One use each	One way of caring for the tool
(i) Builders square	• Used to check if the corners of walls, etc, are accurate right angles (90°)	• Clean and keep it dry • Use it for the right job • Keep in a tool box
(ii) Hand file	• Used to file shoulders of metal	• Clean after use • Use it for the right job

	• Used to file corners of metal	• Keep the teeth sharp • Keep it in a tool box • Oil the metal part
(iii) Rip saw	• Used for cutting a piece of wood into two	• Sharpening • Shaping • Setting • Topping / Jointing • Oiling the blade • Cleaning • Racking the saw
(iv) Gauge rod	• Used to check if the course height is accurate	• Clean and keep it dry • Use it for the right job • Keep in a tool box

3. (a) Explain **each** of the following terms as used in concrete :

(i) fine aggregates

Sand which has been washed and sieved to remove particles greater than 5mm in diameter and is to be used for mortar.

OR

Sieved sand, of diameter less than 5 mm, to be used for mixing mortar.

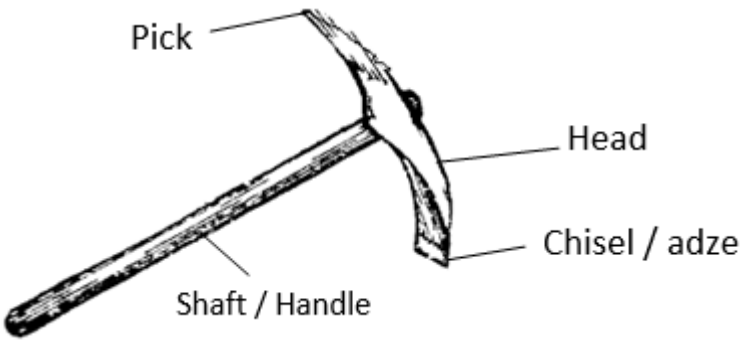
(ii) coarse aggregates

The mixture of sand, stones, gravel and other particles whose diameters are greater then 5 mm and used for mixing mortar

(b) List **three** methods of pictorial drawing

- Oblique drawing
- Isometric drawing
- Perspective drawing

(c) (i) Make a freehand sketch of a pick-axe



(ii) Label any **two** parts on the tool sketched in (c)(i) above

- Shaft / handle
- Head
- Pick
- Chisel / adze

(d) List **two** items made of cast iron.

(e) Draw the symbol for each of the following electronic components

(i) resistor

 or 

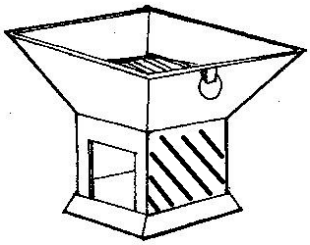
(ii) capacitor



(iii) a switch



(f) The figure below shows a pictorial view of a coal-pot made from a metal plate



Arrange the following steps in a sequential order for making the coal-pot.

(i) File pieces to the required sizes;
(ii) Mark-out the parts of the coal-pot
(iii) Assemble the parts;
(iv) Measure the metal pieces to the required dimensions;
(v) Cut out pieces.

(i) Measure the metal pieces to the required dimensions;
(ii) Mark-out the parts of the coal-pot
(iii) Cut out pieces
(iv) File pieces to the required sizes;
(v) Assemble the parts;

.

4. (a) (i) State the difference between plumbing and levelling in walling

Plumbing refers to checking the vertical uprightness of the wall, whilst
Levelling refers to checking the horizontal levelness of the wall.

(ii) List **two** tools used for plumbing and leveling in walling

Plimb line and Spirit level / bubble level

(b) List **two** examples each of a conductor and an insulator

CONDUCTOR	INSULATOR
Copper	wood
Gold	clay
Silver	plastics
Aluminium	fabric
Molybdenum	paper
Zinc	rubber
Nickel	glass
etc	etc

(c) State **one** purpose each for the following

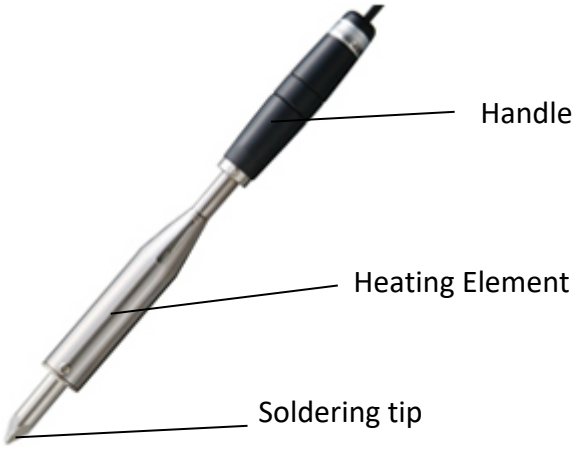
(i) curing a brick;

to make it stronger
to reduce cracking
to reduce shrinking
to make it bond better

(ii) applying a piece of chalk on the face of a file.

to prevent metal chips from clogging the teeth of the file
to prevent pinning
to make its teeth easier to clean

(d) (i) Make a freehand sketch of the soldering iron used in sheet metal work.



(ii) Label any **two** parts on the tool sketched in (d) (i)

- Handle
- Heating element
- Soldering Tip / copper tip

(e) The table below shows two materials to be glued together

Copy and complete the table

Prepared Materials	**Types of adhesive**
(i) Odum to odum	White glue / Polyvinyl acetate (PVA)
(ii) Formica to wawa	Contact adhesive / impact adhesive
(iii) Veneer to odum	Contact adhesive / impact adhesive

April 2011

BASIC DESIGN AND TECHNOLOGY 1
(Pre-Technical Skills 1)

Objective Test
40 minutes

1. Which of the following is a neatening stitch?
A. overcastting
B. feather
C. chain
D. stem

2. Green leafy vegetables and sea foods provide the body with
A. minerals
B. proteins
C. vitamins
D. fats

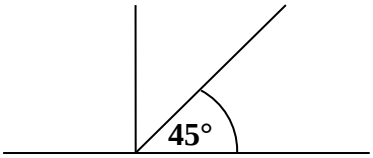
3. Which of the following is a dry method of cooking?
A. boiling
B. baking
C. stewing
D. steaming

4. The lightness or darkness in a drawing is the
A. hue
B. shade
C. tint
D. tone

5. Any two primary colours mixed together in equal proportions produce a
A. complementary colour
B. neutral colour
C. secondary colour
D. tint colour

6. Which of the following lines is used for showing visible outlines?
A. short dashes
B. long chain
C. thick continuous
D. thin continuous

7. Figure 1 illustrates the principle of

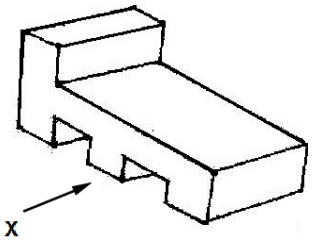


A. oblique axes
B. perspective axes
C. isometric axes
D. orthographic axes

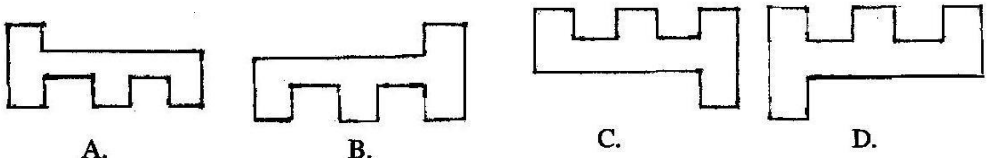
8. From the following options, select the designing media mixed with water
A. charcoal
B. crayon
C. chalk pastel
D. poster colour

9. In design analysis, the size of the unit is considered under
A. cost
B. ergonomics
C. safety
D. function

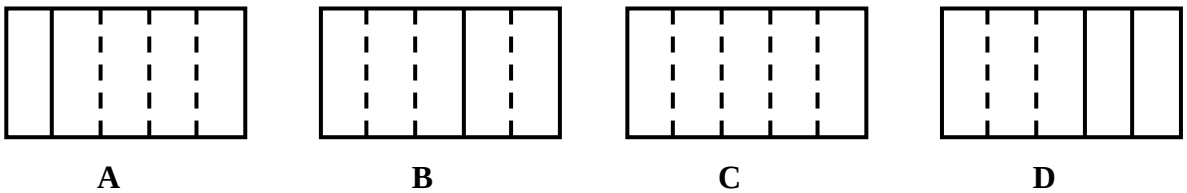
Figure 2 is a block. Use it answer questions 10 and 11



10. Which of the following represents the front elevation in the direction of arrow X?



11. The plan is represented by



12. The two methods of drawing orthographic views are

A. first angle and second angle
B. first angle and third angle
C. second angle and third angle
D. second angle and fourth angle

13. A vertical line is associated with

A. movement
B. rest
C. stability
D. uncertainty

14. Which of the following situations is an indirect risk to a small business?

A. flood
B. fire
C. inflation
D. theft

15. The factors to consider when setting up a sewing business includes

A. capital, location and personal needs
B. capital, location and equipment
C. raw materials, market and business risks
D. raw materials, personal needs and demand

16. Which of the following is not a method of temporary termination of a wall?

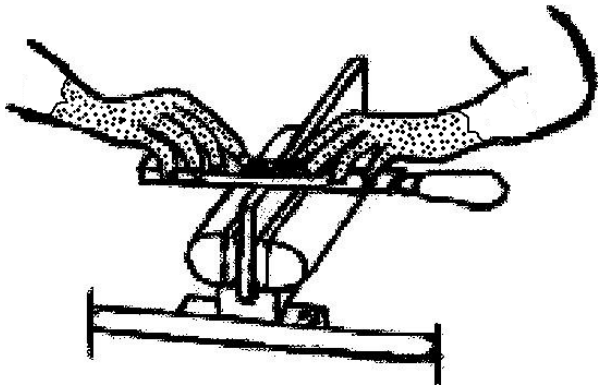
A. toothing
B. racking
C. Indent
D. bonding

17. Which of the following is not a mechanical property of metals?

A. brittleness
B. ductility
C. corrosion
D. malleability

18. The head of a brick hammer is made of

A. high carbon steel
B. mild steel
C. cast iron
D. medium carbon steel



19. The operation shown in the sketch above is

A. cross filing
B. draw filing
C. smooth filing
D. rough filing

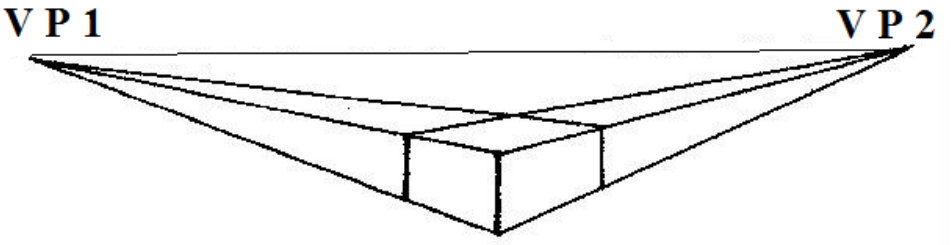
20. The neutral wire has a colour code

A. black
B. red
C. yellow
D. grey

21. Electrical appliances are usually plugged to the electricity mains through

A. switches
B. lamp holders
C. socket outlets
D. ceiling rose

Use figure 3 to answer questions 22 and 23



22. The parts VP1 and VP2 are called

A. viewing points
B. vanishing points
C. viewing planes
D. vanishing planes

23. The object is drawn in

A. oblique
B. isometric
C. one point perspective

D. two point perspective

24. Which of the following is not a fastener?

A. rivets
B. screws
C. soldering bits
D. bolts and nuts

25. The plastic used for making an egg box is

A. acrylic
B. polystyrene
C. nylon
D. polythene

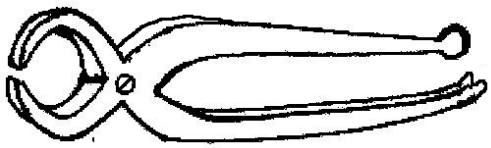
26. Which of the following instruments are most suitable for dividing a circle into equal parts

I. set square
II. protractor
III. a pair of compasses

A. I and II only
B. I and III only
C. II and III only
D. I, II and III

27. Polyvinyl acetate is a type of a/an

A. aggregate
B. adhesive
C. abrasive
D. finish



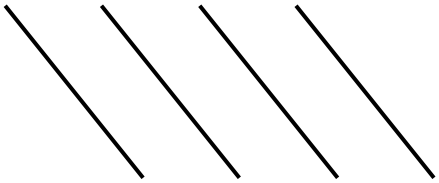
28. The tool shown above is a

A. crowbar
B. pincers
C. screw driver
D. claw hammer

29. To avoid vertical straight joint in a wall, introduce a

A. bat
B. block
C. batten
D. brick

30. The lines drawn below are



A. tangential
B. horizontal
C. vertical
D. inclined

April 2011

BASIC DESIGN AND TECHNOLOGY 1
(Pre-Technical Skills 1)

Objective Test

ANSWERS

1.	A.	overcasting
2.	A.	minerals
3.	B.	baking
4.	D.	tone
5.	C.	secondary colour
6.	C.	thick continuous
7.	A.	oblique axes
8.	D.	poster colour
9.	B.	ergonomics
10.	A.	
11.	A.	
12.	B.	first angle and third angle
13.	C.	stability
14.	C.	inflation
15.	B.	capital, location and equipment
16.	D.	bonding
17.	C.	corrosion
18.	D.	medium carbon steel
19.	B.	draw filing
20.	A.	black
21.	C.	socket outlets
22.	B.	vanishing points
23.	D.	two point perspective
24.	C.	soldering bits
25.	B.	polystyrene
26.	C.	II and III only
27.	B.	adhesive
28.	B.	pincers
29.	A.	bat
30.	D.	inclined

April 2011

BASIC DESIGN AND TECHNOLOGY 2
(Pre-Technical Skills 2)

Essay
1 hour 15 minutes

SECTION A
(CORE SKILLS)
[20 marks]
Answer question 1 [COMPULSORY]

1. (a) State **three** characteristics of a good plain seam.

(b) State **three** causes of food spoilage.

(c) Complete the table below by filling in the blank spaces

Colour	**Association in Nature**	**Symbolism**
Example: Red	Blood	Danger
	Sky	
White		
		Decay
	Past history	Mourning

(d) The headmaster of your school has asked you to design and construct a Kumasi Ventilated Improved Pit (KVIP) toilet for the school

(i) List **three** building materials you will use for making the structure

(ii) List **four** tools for constructing the walls.

SECTION B
[50 marks]

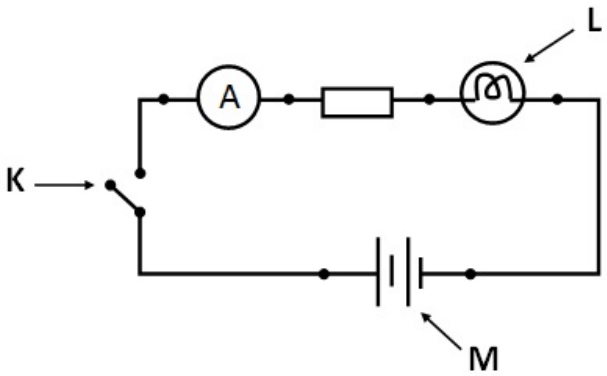
Answer **two** *questions* **only** *from this section*
All questions carry equal marks

2. (a) (i) Explain the term *draw filing*

(ii) Sketch the triangular file

(iii) Label any **two** parts on the sketch in (a)(ii) above.

(b)



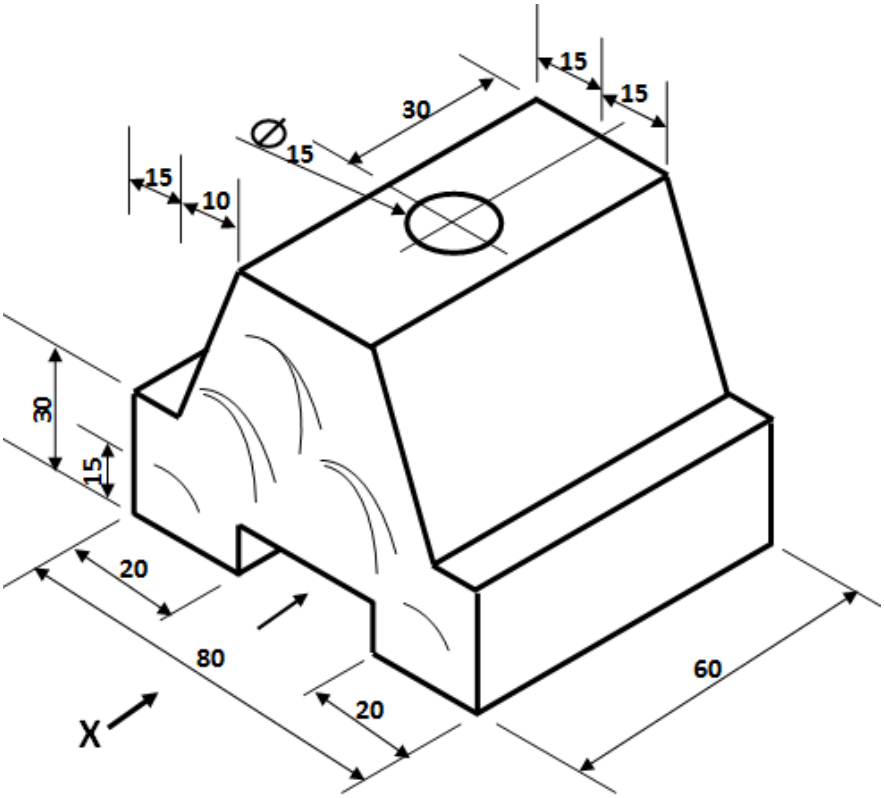
(i) Identify the items labeled **K, L, M** in the circuit diagram above.

(ii) List **two** examples of electrical conductors

(c) (i) State **two** precautions to be taken when cutting with a hacksaw

(ii) List **three** driving tools

3. Figure 1 is a pictorial drawing of a flagstand



(a) Draw full size the following views in first angle projection

(i) front elevation in the direction of arrow X;
(ii) plan

(b) List **one** specific tool for each of the following processes:
(i) marking-out the tenon
(ii) cutting the tenon

(c) State **one** finish for the flagstand

(d) (i) Sketch in pictorial drawing a brick hammer
(ii) Label any **two** parts on the sketch in (d)(i)

4. (a) Explain the following terms:
(i) *pinning*
(ii) *curing*

(b) (i) Make a freehand sketch of an outside calipers
(ii) State **one** use of the outside calipers

(c) State **one** use of the following measuring tools:
(i) steel rule
(ii) tape measure

(d) State **two** methods each for maintaining the tools stated in (c) above.

(e) Copy and complete the table below

Process	One tool used
(i) Plumbing a wall	
(ii) Checking corners of a wall	
(iii) Measuring aggregates	
(iv) Checking courses of a wall	
(v) Picking and spreading mortar	
(vi) Mixing mortar	

April 2011

BASIC DESIGN AND TECHNOLOGY 2
(Pre-Technical Skills 2)

Essay

ANSWERS

1. (a) State three characteristics of a good plain seam.

Firm stitched and strong
Well neatened
Suitable for the fabric.

(b) State three causes of food spoilage.

Enzyme action
Moulds activity
Yeasts activity
Bacteria activity
Insect infestation

(c) Complete the table below by filling in the blank spaces

Colour	Association in Nature	Symbolism
Example: Red	Blood	Danger
Blue	Sky	Peace / Love / Harmony /
White	Cotton, Snow, Wool	Victory / Purity / Righteousness / Innocence
Brown	Earth / Mud	Decay
Black	Past history	Mourning

(d) The headmaster of your school has asked you to design and construct a Kumasi Ventilated Improved Pit (KVIP) toilet for the school

(i) List three building materials you will use for making the structure

- Sand,
- stones,
- iron rods,
- wood,
- cement,
- roofing sheet,
- nails

(ii) List four tools for constructing the walls.

- hand trowel
- spirit level
- float
- gauge rod
- straight edge
- mortar board
- line and pins
- brick hammer / comb hammer
- bolster and club hammer
- builder's square

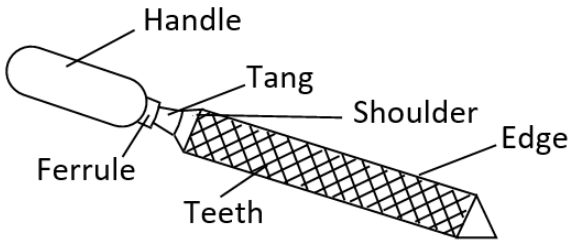
SECTION B
[50 marks]

Answer **two** *questions* **only** *from this section*
All questions carry equal marks

2. (a) (i) Explain the term *draw filing*

A technique used for producing smooth, square edges, especially on pieces of metal. The process works by moving any type of single cut file forwards and backwards along the length of the material's edge

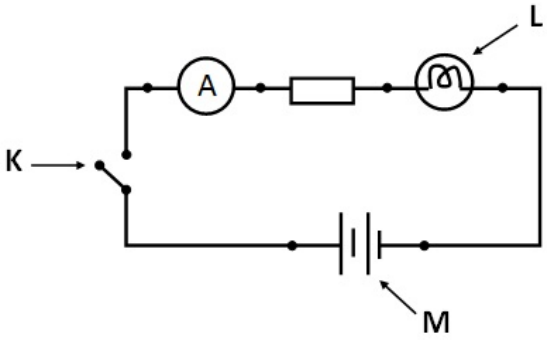
(ii) Sketch the triangular file



(iii) Label any two parts on the sketch in (a)(ii) above.

- Handle
- Tang
- Shoulder
- Ferrule
- Edge
- Teeth

(b)



(i) Identify the items labeled K, L, M in the circuit diagram above.

K - Switch / Key
L - Bulb
M - Battery

(ii) List two examples of electrical conductors

- Copper
- Gold
- Silver
- Aluminium
- Molybdenum
- Zinc
- Nickel
- etc

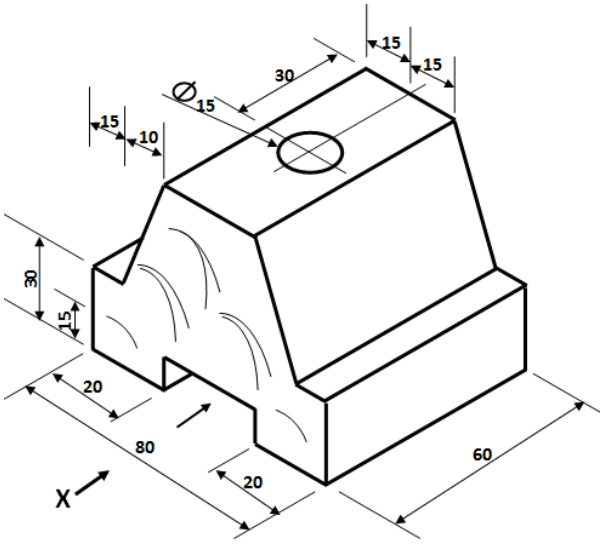
(c) (i) State two precautions to be taken when cutting with a hacksaw

- Tie / cover long or loose hair back to prevent entanglement
- Remove all kinds of jewellery to prevent snagging.
- Wear protective clothing to protect the skin
- Wear protective gloves to protect the hands.
- Use ear protectors such as plugs or mufflers to protect the ear.
- Wear goggles / safety glasses to protect the eyes
- Select correct blade for material being cut
- When cutting, use steady strokes directed away from you.
- Do not apply too much pressure on the blade to prevent breaking
- Make sure the material to be cut is securely fastened

(ii) List three driving tools

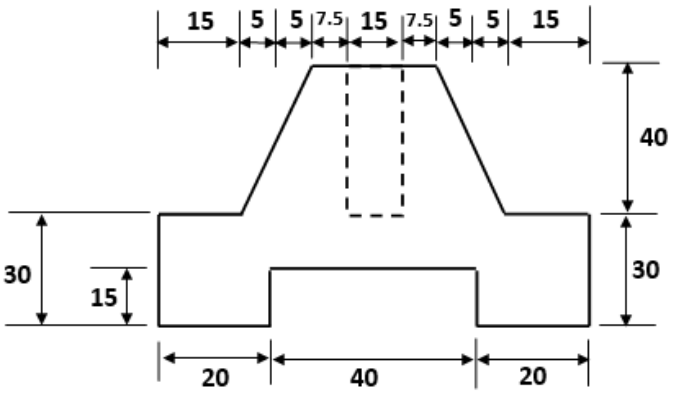
- Claw hammer
- Screw driver
- Nail Punch
- Crow bar / pinch bar
- Tower pincer

3. Figure 1 is a pictorial drawing of a flagstand

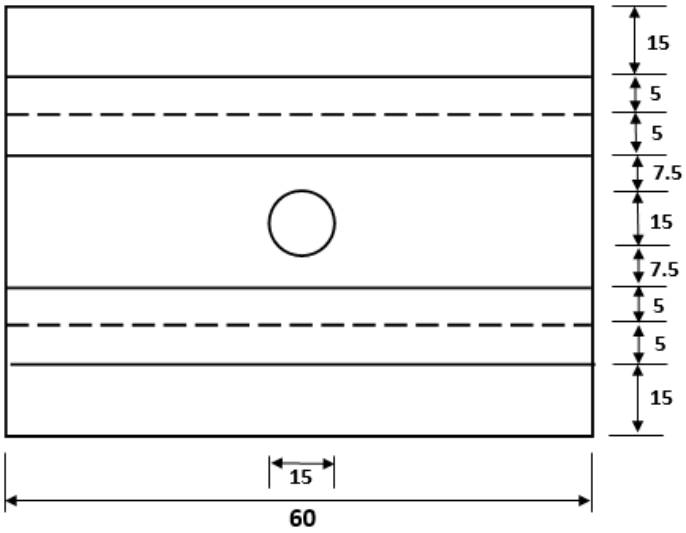


(a) Draw full size the following views in first angle projection

(i) front elevation in the direction of arrow X;



(ii) plan



(b) List one specific tool for each of the following processes:

(i) marking-out the tenon

- Marking knife

- Pencil
- Try-square
- Marking gauge
- Rule / Tape measure

(ii) cutting the tenon

- Try square
- Marking gauge / mortise gauge
- Pencil / Marking knife
- Tenon Saw
- Mallet
- Bevel-edged chisel / firmer chisel

(c) State one finish for the flagstand

- Varnish
- Wax
- Lacquer
- Oil paint

(d) (i) Sketch in pictorial drawing a brick hammer



(ii) Label any two parts on the sketch in (d)(i)

- Striking face
- Handle / shaft
- Chisel
- Head

4. (a) Explain the following terms:

(i) ***pinning***

Pinning refers to the clogging of the **file** teeth with pins, which are material shavings. These pins cause the **file** to lose its cutting ability and can scratch the workpiece.

OR

Pinning occurs when metal pieces, known as pins, get stuck within the teeth of the file. The pins then reduce the effectiveness of the file and also scratch the workpiece.

(ii) ***curing***

Curing of concrete is the provision of adequate moisture, temperature, and time to allow the concrete to achieve the desired properties for its intended use

OR

Curing is the process of maintaining satisfactory moisture content and temperature in freshly cast concrete for a definite period of time immediately following placement

(b) (i) Make a freehand sketch of an outside callipers



OR



(ii) State one use of the outside callipers

Outside callipers are used to measure thicknesses or outside diameters of objects.

(c) State one use of the following measuring tools:

(i) steel rule

- Marking out on a metal plate
- Taking measurements

(ii) tape measure

- Taking measurements
- Setting out
- marking out

(d) State two methods each for maintaining the tools stated in (c) above.

(i) maintaining steel rule

- Keep it clean and dry
- Use it for its intended purpose only
- Avoid bending it
- Avoid scratching its surface

(ii) maintaining tape measure

- After using, roll back into its case
- Keep in a tool box
- Keep it clean and dry
- Use it for its intended purpose only

(e) Copy and complete the table below

Process	One tool used
(i) Plumbing a wall	Plumb line Spirit level
(ii) Checking corners of a wall	Builder's square
(iii) Measuring aggregates	Gauge box
(iv) Checking courses of a wall	Gauge rod
(v) Picking and spreading mortar	Trowel
(vi) Mixing mortar	Shovel spade

April 2010

BASIC DESIGN AND TECHNOLOGY 1
(Pre-Technical Skills 1)

Objective Test
40 minutes

1. Stitches which are removed from an article after serving their purpose are known as
A. joining stitches
B. running stitches
C. temporary stitches
D. permanent stitches

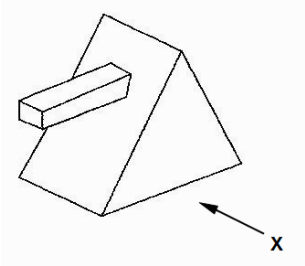
2. What is the common nutrient found in liver and beans?
A. Fats
B. Vitamins
C. Carbohydrates
D. Proteins

3. The **most** popular method of preserving cassava is
A. smoking
B. freezing
C. drying
D. canning

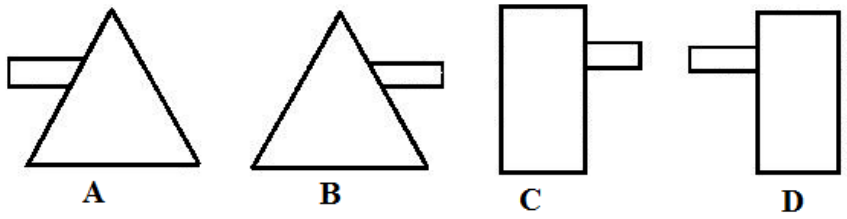
4. The arrangement of shapes in a picture is known as
A. composition
B. drawing
C. blending
D. toning

5. The **most** suitable support for drawing is
A. canvas
B. paper
C. rubber
D. wall

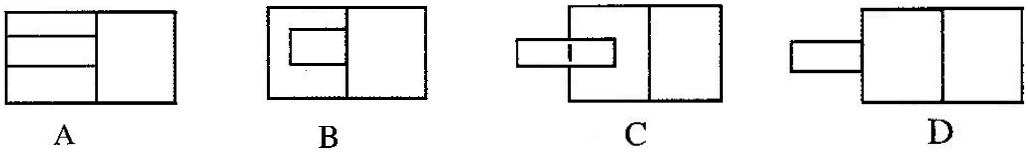
Figure 1 is a block. Use it to answer Questions 6 and 7.



6. Which of the following represents the front elevation of the direction of arrow X?



7. The plan is represented by



8. Graphic designers communicate information that can be

A. seen and heard
B. printed and read
C. seen and read
D. shown and felt

9. The artifact when completed should be able to hold a normal size candle. This specification is classified as

A. function
B. construction
C. ergonomics
D. aesthetics

10. In design, the process used to determine the strengths and weaknesses of an artifact is

A. analyzing
B. testing
C. realization
D. modification

11. The colour code for a live wire is

A. blue
B. red
C. yellow
D. green

12. Which of the following is **not** a basic home preventive maintenance of furniture?

A. Strengthening joints
B. Applying new finish
C. Replacing damaged locks
D. Replacing furniture

13. A method for making several identical copies of an artifact is

A. shading
B. drawing
C. printing
D. painting

14. A drawing of vegetation and buildings is

A. landscape
B. imaginative
C. memory
D. portraiture

15. An indirect risk to a sewing industry is

A. low quality products
B. misuse of profits
C. inflation controls
D. increasing capital

16. The snips is used for

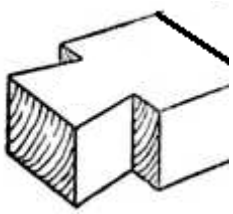
A. striking
B. holding
C. measuring
D. cutting

17. Which of the following tools has a lever cap?

A. Bench vice
B. Hacksaw
C. Outside calipers
D. Jack plane

18. The term laying is used in

A. block work
B. wood work
C. metal work
D. plastic work



19. The sketch shown above is a

A. tongue
B. tenon
C. mortice
D. dovetail

20. Which of the following electronic components allows current to flow in only one direction?

A. Inductor
B. Diode
C. Switch
D. Capacitor

21. Which of the following applies to a two-point perspective drawing?

A. Angle of projection is 30°
B. Angle of projection is 60°
C. Object is viewed from a corner
D. Front view is drawn first

22. Which of the following is a physical property of metals?

A. Weight
B. Brittleness
C. Ductility
D. Malleability

23. The body of a spoke shave is made of

A. cast iron
B. mild steel
C. copper
D. lead

24. Which of the following materials is a ferrous metal?

A. Lead
B. Steel
C. Copper
D. Aluminium

25. The most appropriate surface finish for a garden chair is

A. wax polish
B. emulsion paint
C. oil paint
D. sanding sealer

26. One purpose of using a flux in soldering is to

A. prevent formation of oxides
B. wash surfaces of the work
C. heat the soldering iron

D. make the joint neat

27. A fuse wire breaks when there is
A. high voltage
B. excess current
C. low voltage
D. low current

28. Which of the following is the most appropriate instrument for drawing objects in oblique?
A. 30° set square
B. 45° set square
C. Ruler
D. A pair of dividers

29. In orthographic projection, the plan is projected onto the
A. vertical plane
B. horizontal plane
C. side plane
D. auxiliary plane

30. Which of the following is a thermoplastic material?
A. Polyester resin
B. Phenol formaldehyde
C. Polyvinyl chloride
D. Melamine formaldehyde

April 2010

BASIC DESIGN AND TECHNOLOGY 1
(Pre-Technical Skills 1)
Objective Test
ANSWERS

1.	C.	temporary stitches
2.	D.	Proteins
3.	C.	drying
4.	A.	composition
5.	B.	paper
6.	A.	
7.	A.	
8.	C.	seen and read
9.	A.	function
10.	B.	testing
11.	B.	red
12.	D.	Replacing furniture
13.	C.	printing
14.	A.	landscape
15.	C.	inflation controls
16.	D.	cutting
17.	D.	Jack plane
18.	A.	block work
19.	D.	dovetail
20.	B.	Diode
21.	D.	Front view is drawn first
22.	A.	Weight
23.	A.	cast iron
24.	B.	Steel
25.	C.	oil paint
26.	A.	prevent formation of oxides
27.	B.	excess current
28.	B.	45° set square
29.	B.	horizontal plane
30.	C.	Polyvinyl chloride

April 2010

BASIC DESIGN AND TECHNOLOGY 2
(Pre-Technical Skills 2)

Essay
1 hour 15 minutes

SECTION A
(CORE SKILLS)
[20 marks]
Answer question 1 [COMPULSORY]

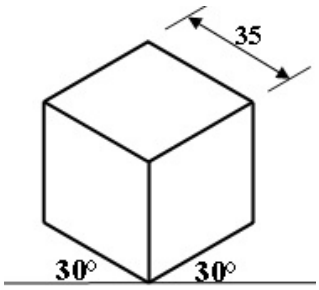
1. (a) What is a *seam*?

(b) Suggest the correct stitches for

(i) fringed edge of a tablecloth

(ii) making of a baby's petticoat

(c) The figure below is a cube drawn in isometric



Sketch the cube in **two-point** perspective drawing

(d) (i) What is *texture*?

(ii) List the **two** types of texture

(e) Copy and complete the table below by providing the type of texture associated each object listed

OBJECT	TEXTURE
Banana	
Pineapple	
Unwashed terrazzo	

SECTION B
[50 marks]

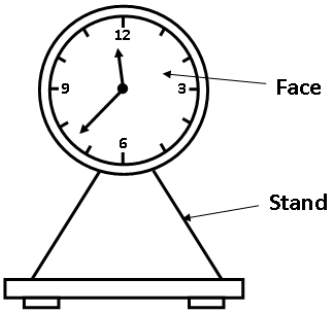
Answer **two** *questions* **only** *from this section*
All questions carry equal marks

2. (a) State two methods suitable for joining sheet metals permanently

(b) Sketch a lap joint used in sheet metalwork

(c) List three items that can be manufactured from sheet metal

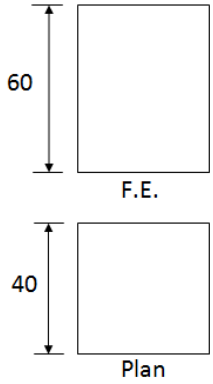
(d) Figure 1 shows the front view of a clock. Use it to answer the questions that follow.



(i) State two properties of acrylic that makes it suitable for making the face of the clock.

(ii) List two tools that can be used to mark-out the shape of the face of the clock.

3. Figure 2 shows the front elevation and plan of a square prism. Use it to answer questions 3(a)



(a) Draw full size, the following:
- (i) front elevation
- (ii) plan
- (iii) surface development

(b) State one use of each of the following electronic components:
- (i) capacitor
- (ii) inductor
- (iii) diode

(c) Make a freehand pictorial sketch of a wooden mallet.

4. (a) Make a freehand sketch of a cross-cut saw

(b) List three materials from which sandcrete blocks are manufactured

(c) State one reason each for using the following in the workshop
- (i) apron;
- (ii) gloves;
- (iii) helmet;
- (iv) goggles

April 2010

BASIC DESIGN AND TECHNOLOGY 2
(Pre-Technical Skills 1)

Essay

ANSWERS

1. (a) What is a *seam*?

A seam is a line of stitching, joining two or more pieces of fabric together permanently.

or

a line where two pieces of fabric are sewn together in a garment or other article

(b) Suggest the correct stitches for

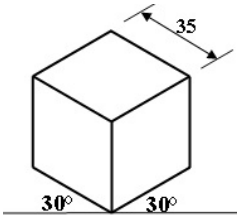
(i) fringed edge of a tablecloth

Neatening stitches, such as overcasting, blanket stitch and hemming stitch

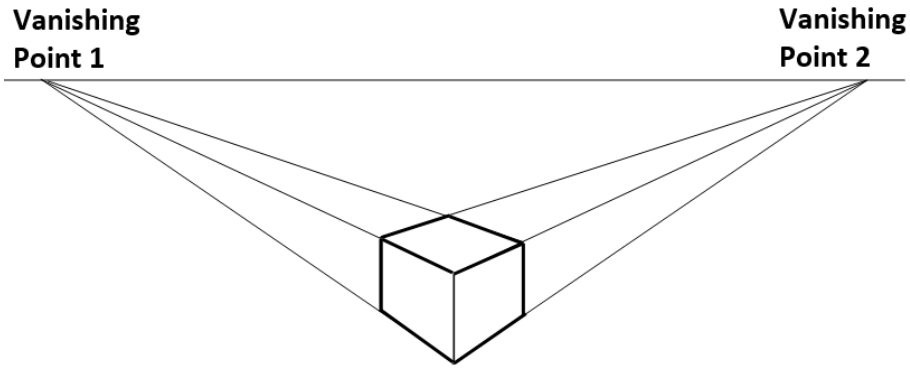
(ii) making of a baby's petticoat

Decorative stitches, such as chain stitch, cross stitch, stem stitch and blanket stitch

(c) The figure below is a cube drawn in isometric



Sketch the cube in **two-point** perspective drawing



(d) (i) What is *texture*?

The feel or appearance of surface or substance

Or

How rough or smooth a surface is or appears to be

(ii) List the two types of texture

Visual (virtual) texture

Tactile / Actual (tangible) texture

(e) Copy and complete the table below by providing the type of texture associated each object listed

OBJECT	TEXTURE
Banana	Tactile / Smooth
Pineapple	Tactile / Rough
Unwashed terrazzo	Visual / rough

SECTION B
[50 marks]

Answer **two** *questions* **only** *from this section*

All questions carry equal marks

2. (a) State two methods suitable for joining sheet metals permanently

Soldering

Welding

Riveting

Brazing

(b) Sketch a lap joint used in sheet metalwork

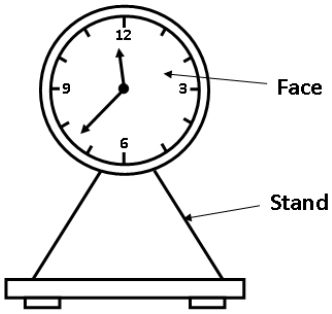


(c) List three items that can be manufactured from sheet metal

- Roofing sheets
- Barrels
- Jewellery
- Graters
- Cooking utensils
- Tins
- Cans
- funnels
- buckets

- machine parts
- Water reservoir

(d) Figure 1 shows the front view of a clock. Use it to answer the questions that follow.



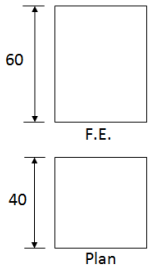
(i) State two properties of acrylic that makes it suitable for making the face of the clock.

- Excellent optical clarity
- Excellent resistance to sunlight
- Rigid, with good impact strength
- Flexible and shock resistant.
- Easily cut, corrosion resistant and a good insulator.

(ii) List two tools that can be used to mark-out the shape of the face of the clock.

- Centre punch
- Felt-tip pen / marker pen
- Try square
- Pair of dividers
- Pencil
- Template

3. Figure 2 shows the front elevation and plan of a square prism. Use it to answer questions 3(a)

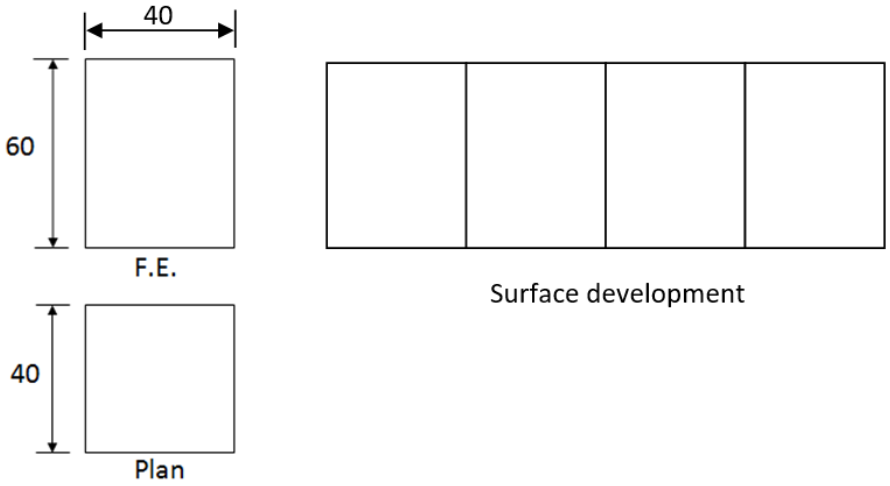


(a) Draw full size, the following:

(i) front elevation

(ii) plan

(iii) surface development



(b) State use one of each of the following electronic components:

(i) capacitor

- Stores electrical current for future use
- Used to operate the tuning dial of a radio
- Used in loudspeakers to block DC and allow only AC to flow
- Used to smooth out AC currents

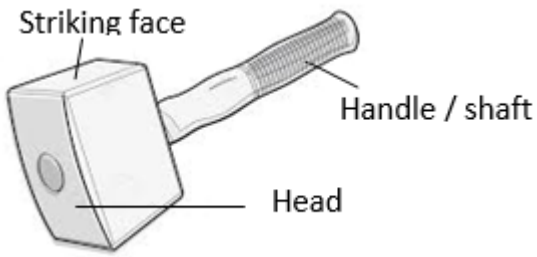
(ii) inductor

- Used in electrical circuits to block AC while allowing DC to pass
- Used to create a filter by blocking high frequency current and allowing only a certain frequency to pass through the circuit
- Used as energy storage devices in transformers, motors and computers
- Used in proximity sensors, such as those used in traffic lights

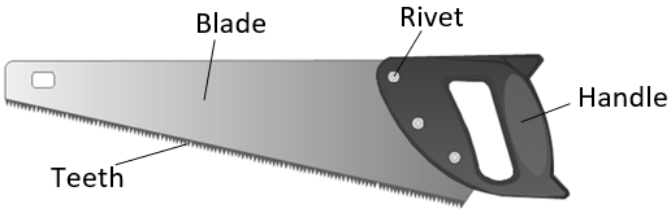
(iii) diode

- Used as rectifiers - to convert AC to DC by allowing electric current to flow in only one direction
- Diodes are also used in/for the following:
  - signal limiters,
  - voltage regulators,
  - switches,
  - signal modulators,
  - signal mixers,
  - signal demodulators,
  - oscillators

(c) Make a freehand pictorial sketch of a wooden mallet.



4. (a) Make a freehand sketch of a cross-cut saw



(b) List three materials from which sandcrete blocks are manufactured

- Sand,
- Cement
- Lime

(c) State one reason each for using the following in the workshop

(i) apron;

To prevent clothes from becoming dirty / stained.

(ii) gloves;

To protect the hands from injury and dirt.

(iii) helmet;

To protect the head from injury

(iv) goggles

To protect the eyes from injury / prevent particles from entering the eyes